



về "đường cong học tập" (learning curve / learning rate) và mức giảm giá 20% quy mô lớn so với 20% giai đoạn đầu (trang 77)

Tránh phát thải 923 triệu tấn CO₂: trang 66

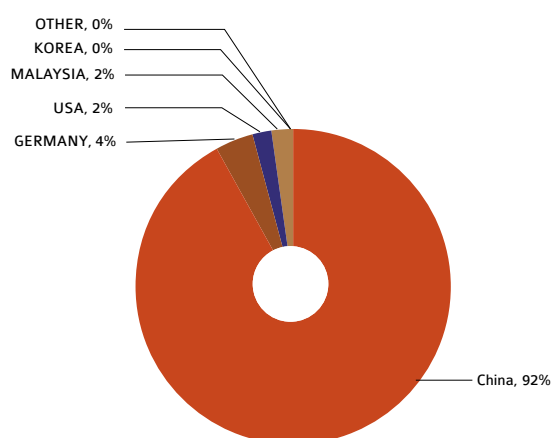
Giá module giảm xuống 0,115 USD/W (2023) và dưới 0,10 USD/W (2024) do dư thừa công suất: trang 55 - 79

In China, five new companies entered into polysilicon production and Chinese total manufacturing capacity reached 2.3 million tons/year, nearly double the previous year (1.16 million tons/year). China produced 1.47 million tons of polysilicon in 2023 and is expected to increase to 1.6 million to 1.8 million tons in 2024. According to the China Photovoltaic Industry Association, CPIA, China produced 1.06 million tons of polysilicon in the first half of 2024, a 60.6% increase on the previous period. Due to the continuous trends of lowering prices, some companies reconsidered capacity expansion and halted operation to ease the demand gap. Figure 4.2 shows the share of polysilicon production by country.

IEA PVPS member countries producing polysilicon are China, Germany, the USA, Malaysia, South Korea, Norway and Japan. Chinese production share in 2023 increased to 92% from 86% in 2020.

Because of the rapid growth of manufacturing capacity and the gaps between demand and supply continuing, the price of polysilicon dropped from 22.13 USD/kg at the end of 2022 to the 7 USD/kg range by the end of June 2023, and it remained below 10 USD/kg thereafter. In January 2024, the price level was 7 USD/kg and it further dropped to 4.36 USD/kg in June 2024. There are 17 polysilicon companies in China, but as of the end of May, at least nine of them have reportedly stopped operations at their factories for maintenance or other reasons.

FIGURE 4.2: SHARE OF PV POLYSILICON* PRODUCTION IN 2023 BY COUNTRY



*Production for semiconductors is included

Source: IEA PVPS, RTS Corporation

Consumption unit of polysilicon per W of solar cells dropped from 2.7 g/W in 2021 to 2.3 g/W in 2022. In 2023, it further dropped to 2.2g /W. Consumption per unit is declining due to the improvement of solar cell efficiency, thinner and larger wafers, and development of slicing technologies.

Most of the major polysilicon manufacturers use the Siemens process, conceived as a manufacturing process of polysilicon for the semiconductor industry. It is estimated that the Siemens process polysilicon accounted for 84% of the total production (1.35 million tons). Production of polysilicon from FBR process increased and accounted for 16.1%.

In 2021, the USA decided to ban imports of material from China-based Hoshine Silicon Industry that produces metallic silicon, a base material of polysilicon. Then, in December 2021, the Uyghur Forced Labor Prevention Act (UFLPA) was enacted. On June 21, 2022 the USA started to enforce the ban on the importation of goods that were manufactured wholly or in part with forced labor in China, especially from the Xinjiang Uyghur Autonomous Region. This measure combines with the IRA that provides tax credits for polysilicon production (3 USD/kg). In the USA, several production plans were reported. REC Corporation decided to start reoperation of polysilicon plants with the investment guaranteed by Hanwaha Solution of South Korea. Highland Materials, a USA manufacturer plans to set up a 16 000 ton/year solar-grade polysilicon plant. As was mentioned in the previous section, polysilicon production is planned in India, Australia. In the Middle East, several new manufacturing projects are reported. GCL Technology has announced a partnership with Mubadala Investment, a state-owned company of the United Arab Emirates (UAE), to establish the Middle East's first polysilicon plant in June 2024. United Solar Polysilicon (FZC) has broken ground on a polysilicon plant with a metallurgical process in SOHAR Port and Freezone, Oman. The current lower price level of polysilicon might affect the progress of these projects and China's dominant share is expected to continue for the time being.

INGOTS & WAFERS

To produce sc-Si ingots or mc-Si ingots, the basic input material consists of highly purified polysilicon. The ingots need to be cut into bricks or blocks and then sawn into thin wafers. In the past two types of silicon ingots were manufactured, sc-Si ingots and mc-Si ingots, but sc-Si has become the industry standard and had an almost 100% share in 2023 according to the International Technology Roadmap for Photovoltaic (ITRPV) 2024.

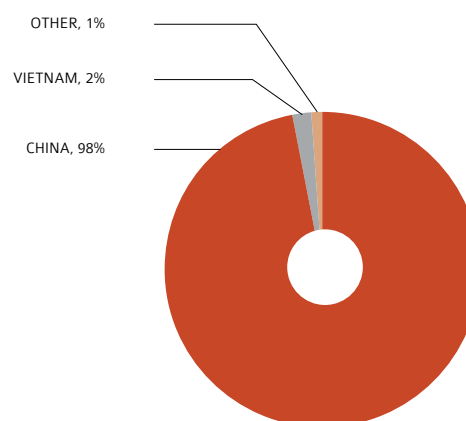
Global wafer production amounted to 682 GW in 2023, about a 79% increase from 381 GW in 2022. 291 GW/year of new capacity was added in 2023 and the global production capacity of wafers as of the end of 2023 increased from 683 GW/year in 2022 to 974 GW/year. Production capacity is highly concentrated in China, with an

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almost 98% share. Other major countries manufacturing wafers are Vietnam (18.5 GW/year), Malaysia, and Norway.

As shown in Figure 4.3, China has more than 98% of the global production of wafers. In 2023, China produced 668 GW of wafers. Of this total, around 70 GW of wafers were exported to other solar cell manufacturing countries such as Vietnam, Malaysia, Thailand, Singapore, Chinese Taipei and India. In Vietnam and Malaysia, major Chinese solar cell manufacturers have production capacity. In this segment, the Top 10 companies are all Chinese and combined they have a capacity of 831 GW/year, or 85.5% of the world's total production capacity. They produced approximately 668.3 GW, accounting for 98.1% of the world's production

FIGURE 4.3: SHARE OF PV WAFER PRODUCTION - 2023 BY COUNTRY



SOURCE: EIA PVPS, RTS Corporation

TABLE 4.1: PRODUCTION CAPACITY AND PRODUCTION AMOUNT OF TOP 5 WAFER PRODUCERS IN 2023

RANK	COMPANY	FACTORY LOCATION	PRODUCTION CAPACITY (GW/YEAR)	PRODUCTION (GW)
1	TCL ZHONGHUAN	CHINA	155	133.7
2	LONGI GREEN ENERGY TECHNOLOGY	CHINA	167.4	124.9
		MALAYSIA	2.6	2.6
3	JINKOSOLAR	CHINA	78	69
		VIETNAM	7	7
4	GCL GROUP	CHINA	58.5	51.1
5	JA SOLAR TECHNOLOGY	CHINA	78.5	45.4
		VIETNAM	5	4.7

SOURCE: EIA PVPS, RTS Corporation

In addition to major ingot/wafer manufacturers, some PV cell/module manufacturers also manufacture silicon ingots and wafers for their in-house uses. There was a trend for some of these major PV module manufacturers that established vertically integrated manufacturing to shift to procuring wafers from specialized manufacturers because of cost and quality advantages under the cost pressure. However, recently, some major companies have increased their in-house production capacity to reduce the risks associated with procurement. Table 4.1 shows the production capacity and production amount of Top 5 wafer producers

In 2023, the market share of n-type and p-type single crystalline Si was 30% and 70% respectively. It is expected that n-type will rapidly increase to approximately 63% in 2024. In 2023, market share of 182 mm (M10, square or rectangular) was 78% and 210 mm (G12) had a 20% of the total production. Share of 166 mm sized wafer decreased to less than 2% as large-sized wafers become the

mainstream products. Wafer thickness further decreased in 2023. Average wafer thickness for p-type products in 2023 was 150 μm down by 5% in 2022. The average thickness of n-type wafers for TopCon was 125 μm . n-type is expected to be thinner and reach the 100 μm level in the short term. For cost reduction reasons, Tungsten wire is used as an alternative to diamond wire to allow thinner wafers.

The spot price of c-Si wafers generally follows the price of polysilicon. The price of 182 mm sc-Si wafers was 50.3 USD cents/piece in December 2022 and the trend down continued into 2023 to reach a spot price at the end of December 2023 of 34.9 USD cents/piece. This price decline was caused by the oversupply of wafers while the production capacity of large solar cells did not increase as much as that of wafers. In early 2023, short supply of quartz sand, the raw material for crucibles used in the production of ingots was reported, however process improvements meant it was no longer an issue in 2024.



In the first half of 2024, China produced 402 GW of wafers, a 58.6% increase on the same period of the previous year. Of this, 38.8 GW was exported to mainly ASEAN countries, where major PV manufacturers have solar cell manufacturing facilities. It is reported that more than half of the wafer manufacturing companies dropped operating intensity or halted production to narrow the supply and demand gap. Considering over capacity, plans for about 15 GW/year of new capacity were canceled or postponed.

Outside of China, plans for new and expanded wafer production capacity were reported in 2023. In India, several companies are planning vertical production from wafers to PV modules using the subsidy under the Production Linked Incentive (PLI) scheme. It is reported that total wafer capacity in India will reach 56 GW/year by 2026. According to the USA's Solar Energy Industries Association, SEIA, a total 20 GW/year wafer production capacity was announced in the USA. A Norwegian company, NorSun is planning to have 5 GW/year capacity in Oklahoma (USA) and Hanwha is building a 3.3 GW/year ingot/wafer plant in Georgia (USA). NexWafe (Germany) that is building a 250 MW/wafer plant in Bitterfeld, Germany announced a plan to establish a 6 GW/year factory in the USA. However, it is unclear whether these plans will come to fruition due to the over-supply issues, and some reports of cancelled projects have already been reported - Cubic PV cancelled its project of 10 GW/year in the USA. Given these circumstances, there is no doubt that China will remain the primary wafer supplier for the time being.

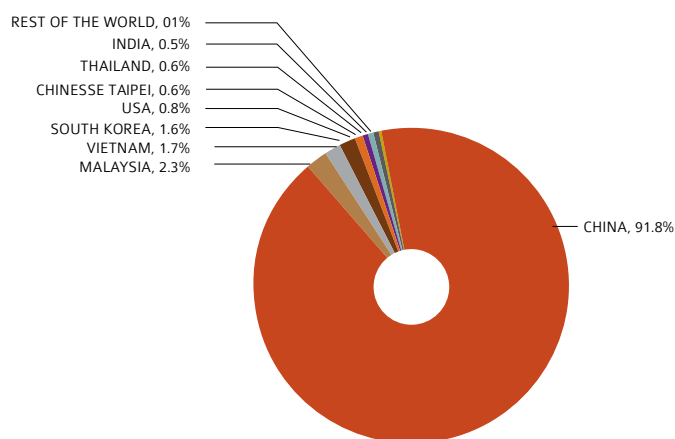
SOLAR CELL AND MODULE PRODUCTION

Solar Cell Production

Global solar cell (c-Si and thin-film solar cell) production in 2023 reached 644 GW, a 63% increase from 394 GW in 2022. Global solar cell manufacturing capacity was 1 032 GW/year as of the end of 2023, as for the first time it passed the 1 TW/year level.

China's dominant position was further strengthened in this sector as China produced 591 GW of solar cells, and increased its share of global production to 91.8% from 84% in 2022 as shown in Figure 4.4. China's solar cell manufacturing capacity was 930 GW/year at the end of 2023 with 424 GW/year of newly added capacity. In China, 310 GW of solar cells were produced in the first half of 2024, a 38.1% increase from the same period of 2023. Major countries other than China that reported production of solar cells in 2023 are Malaysia, Vietnam, Thailand, South Korea, India but also the USA and Japan. Figure 4.4 shows the share of solar cell production by country in 2023. Production in Vietnam, Malaysia and Thailand has been increasing as demand for modules exempt from the USA's antidumping duties (AD) and countervailing duties (CVD) imposed on Chinese products grows. USA imported about 3.5 GW of solar cells from these countries. In the USA, solar cell production is mainly conducted by First Solar with CdTe thin-film PV technology.

FIGURE 4.4: SHARE OF PV CELL PRODUCTION - 2023 BY COUNTRY



SOURCE IEA PVPS, RTS CORPORATION

TABLE 4.2: GLOBAL TOP FIVE MANUFACTURERS IN TERMS OF PV CELL/MODULE PRODUCTION AND SHIPMENT VOLUME (2023))

RANK	SOLAR CELL PRODUCTION (GW)		PV MODULE PRODUCTION (GW)		PV MODULE SHIPMENT (GW)	
1	TONGWEI SOLAR	80.8	JINKOSOLAR	83.5	JINKOSOLAR	78.5
2	JINKOSOLAR	63.9	LONGI GREEN ENERGY TECHNOLOGY	72.8	LONGI GREEN ENERGY TECHNOLOGY	67.5
3	LONGI GREEN ENERGY TECHNOLOGY	62.3	JA SOLAR TECHNOLOGY	60	TRINA SOLAR	65.2
4	JA SOLAR TECHNOLOGY	45.5	TRINA SOLAR	58.9	JA SOLAR TECHNOLOGY	55.3
5	TRINA SOLAR	44.3	CANADIAN SOLAR	31.4	TONGWEI GROUP	31.1

NOTE: PRODUCTION VOLUMES ARE MANUFACTURERS' OWN PRODUCTION, WHEREAS SHIPMENT VOLUMES INCLUDE COMMISSIONED PRODUCTION AND OEM PROCUREMENT

SOURCE IEA PVPS, RTS CORPORATION

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As shown in Table 4.2, the Top five solar cell manufacturers are Chinese companies.

In 2023, p-type PERC technology had the largest market share of about 64%, but a quick transition to n-type is expected to advance in 2024. According to CPIA, the share of n-type manufacturing capacity for TOPCon, silicon heterojunction solar cell (SHJ), back contact type (IBC), etc. is expected to reach 69% by the end of 2024. The share of TOPCon technology increased from 10% in 2022 to about 30% in 2023 and it is expected to increase to about 50% in 2024 and become the mainstream technology. SHJ's share in 2023 was 5% and IBC's share was 2%. The two principal technological developments that manufacturers are investing in is to achieve higher efficiency or to reduce silver (Ag) usage. According to the Silver Institute, demand for Ag for solar cells was up 64% from the previous year to 193.5 million ounces in 2023 and accounted for 16% of total silver demand (1.195 billion ounces).

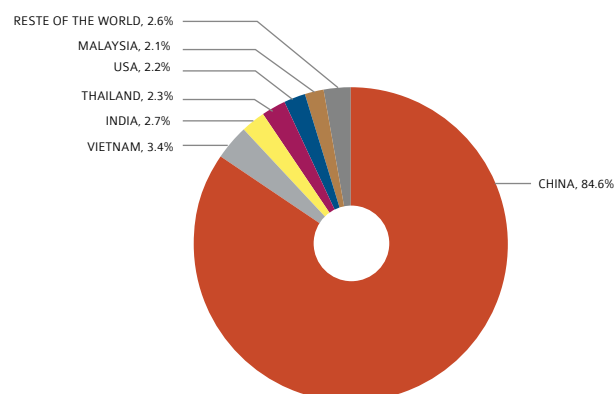
The spot price of solar cells has been changing, following wafer prices and the supply and demand conditions in the market, and the cell manufacturing segment is also affected by the demand and supply gap. In January 2023, the spot price of 182 mm solar cells was 10 USD Cents/W. It had changed according to the demand and p-type products dropped to around 4 USD cents/W in December 2023. N-type products (182 mm) in the same period had a higher price than p-type at 5.6 USD cents/W.

A diversification of solar cell manufacturing locations is expected in the future. In India, cell manufacturing capacity is expected to be more than 75 GW/year by the end of 2026. In the USA, a total of 43 GW/year of new capacity addition has been announced since the IRA was passed. In addition, plans to build solar cell manufacturing bases or capacity additions have also been reported in European countries such as Italy, France as well as Indonesia, Australia, Laos, Brazil, but also in the Middle East and Africa. As was mentioned before, given the current price and demand supply gap, it is uncertain if these plans will eventuate or not.

PV MODULE PRODUCTION

As well as solar cells, global PV module manufacturing capacity surpassed 1 TW/year in 2023 with 1 032 GW/year for the first time as 419 GW/year of new capacity was added. Global PV module production (c-Si PV module and thin-film PV module) reached 612 GW, an almost 62% increase from 324 GW in 2022. As shown in Figure 4.5, China continued to be the largest producer of PV modules in the world. China produced 510 GW of PV modules in 2023 from its 920 GW/year of production capacity. China's production share showed growth from 77.8% in 2022 up to 84.6%. In 2023, China recorded the highest exported amount of PV modules at 211.7 GW, a 37.8% increase from the previous year. As shown in Table 4.2, the Top five PV module manufacturers are Chinese companies. In the first half of 2024, China produced 271 GW of PV modules, up 32.8% year-on-year. Of this, 129.2 GW was exported. The export volume increased 19.7% year-on-year, while export value decreased due to price drops. Because of the rapid increase of supply capacity, most PV module producers have been facing thinner margins so that a reorganization of PV module players has started in China. Announced capacity additions and new entry business plans in China were cancelled or postponed because of the huge oversupply situation.

FIGURE 4.5: SHARE OF PV MODULE PRODUCTION IN 2023 BY COUNTRY



SOURCE IEA PVPS, RTS CORPORATION



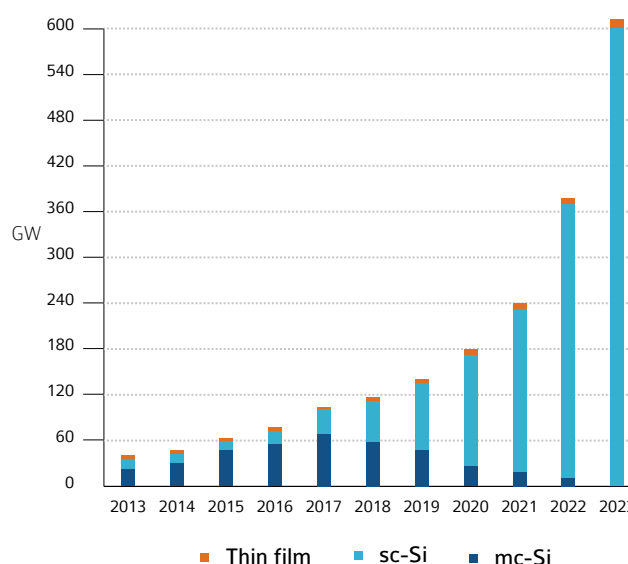
In 2023, the second largest PV module producing country was Vietnam (21 GW) followed by India (16.6 GW). Other countries producing PV modules are Thailand, Malaysia, USA, Canada, South Korea, Chinese Taipei, Türkiye, Singapore and some European countries are listed - but production amounts are smaller than those countries. In India, according to the Ministry of Renewable Energy (MNRE), PV module production capacity reached 64.5 GW/year and it could be 150 GW/year by the end of 2026. The Indian government promotes domestic production by introducing its certification program (ALMM) and Production Linked Incentive (PLI). In the USA, under its policy to guard domestic manufacturers with import duties and incentives under IRA, PV module production increased from 7.3 GW in 2022 to 13.5 GW in 2023, and 20 GW/year of new capacity was added by eight companies for a total manufacturing capacity of 35 GW/year. Further capacity increases are expected in the USA. Under the IRA, a total of 85 GW/year of new PV module manufacturing facilities have been announced. Plans for the regional production of PV modules are becoming active in Europe as well, driven by the demand for lower carbon footprint modules, economic stimulation, and to mitigate the risk of depending on specific production locations under the initiative by the European Union (EU). The Net Zero Industry ACT (NZIA) enforced in June 2024 is expected to promote local production. However, several European companies decided to enhance their capacity in USA instead of Europe due to more favorable conditions.

As well as other products in the supply chain, PV module price was affected by the supply and demand gap and increasing inventories in the market. In January 2023, the average spot price of 182 mm PERC PV modules with a nominal power of 550 W to 595 W was 0.223 USD/W. The price dropped to 0.187 USD/W in June then gradually decreased to 0.115 USD/W at the end of 2023. While dropping prices stimulate the demand, PV module manufacturers are facing difficult situations. One of the major PV module manufacturers requested the Chinese government crackdown on companies selling at unfair prices and review the bidding rules which use price as the only selection criteria. It is reported that manufacturers are now exposed to the risk that product quality will decline if cost reductions are further prioritized, effectively creating a risk to final clients with regards to safety, reliability, and performance throughout the entire operational period of the PV modules. Because of the very competitive environment over prices, major companies are shifting their business to solutions or system integration businesses combining PV products with their inhouse batteries or inverters.

PV MODULE TECHNOLOGY

In 2023, 599.7 GW of c-Si PV module was produced. As shown in Figure 4.6, the share of c-Si PV module was 98%, a slight increase from 97.5% in 2022. It should be noted that the transition to sc-Si PV modules advanced in 2023, and mc-Si disappeared from the market. As mentioned in the section on wafers and cells, adoption of larger-sized solar cells further increased. For the utility-scale segment, 70% of the PV modules used had a surface area of 2.5 m² to 3.0 m² in 2023. In this segment, bifacial PV modules are also commonly used. While larger sized PV modules contribute to a reduction of the LCOE, the heavier weight sometimes becomes an issue as it increases the cost of trackers and construction. PV modules adopting half-cut c-Si solar cells are now mainstream, accounting for more than 90% of all the PV modules in 2023. PV modules with third-cut cells and further separated cells are manufactured as well. Shingled PV module technology (overlapping the edges of solar cells without ribbons) and seamless soldering technology were also adopted.

FIGURE 4.6: PV MODULE PRODUCTION PER TECHNOLOGY IN IEA PVPS COUNTRIES 2013 - 2023



SOURCE IEA PVPS, RTS CORPORATION

THE UPSTREAM PV SECTOR / CONTINUED

Thin film PV production in 2023 was 12.5 GW and the share of thin-film PV slightly decreased from 2.5% in 2022 to 2% in 2023. The majority of the thin-film PV modules were CdTe PV modules produced by First Solar (USA). First Solar shipped 11.4 GW of PV modules produced with 16.6 GW/year of capacity in the USA and Malaysia. China had 540 MW/year of CdTe capacity with 3 companies, but production scale remains low in comparison to First Solar. Other thin-film technologies produced in 2023 were CIGS and amorphous-silicon PV modules.

EMERGING TECHNOLOGY

In several IEA PVPS member countries such as China, Japan, South Korea, USA, Germany, the Netherlands and Sweden, R&D efforts on emerging PV technologies are underway. In particular, since the conversion efficiency has been rapidly improved in a short period of time, efforts toward the practical application of perovskite technology have been continued in 2023. Perovskite solar cells are expected to have the potential for high conversion efficiency, low material costs, and a low carbon footprint process. Approximately 70 to 80 companies around the world are working on development with the aim of commercializing the manufacture of single-junction perovskite solar cells and tandem solar cells that combine perovskite solar cells with other technologies.

Several companies in China and Europe are operating pilot production lines. As of March 2024, there are two manufacturers commercially producing and selling perovskite solar cells: Hangzhou Microquanta Semiconductor of China, which produces flat glass modules, and Saule Technologies of Poland, which mainly produces perovskite solar cells for consumer products (such as for electronic shelf power supplies). Several other companies have begun shipping samples. Most companies, including those currently preparing for production, plan to begin commercial production around 2024 to 2026, after that some of those companies have plans to expand production capacity to GW/year level.

In China, about 20 companies have announced plans to produce perovskite solar cells. In the “Action Plan for Innovation and Development of the Smart Photovoltaic Industry (2021-2025)” announced in 2021, the Ministry of Industry and Information Technology of China (MIIT) announced a policy to promote the smartification of the entire PV industry. In this regard, advance research and industrialization of new solar cell modules, such as perovskite and tandem types are promoted. Development and application of building-integrated photovoltaics (BIPV) with long service life and higher safety is also promoted. Under these policies, glass-glass perovskite modules produced by the pilot lines have begun to be installed on roofs and walls of buildings as a demonstration project. Some companies have announced that they have passed the IEC 61215/61730 certification test. For example, in January 2023, Hangzhou Microquanta Semiconductor of China announced that its mass-produced perovskite solar cell module, “Microquanta Module- α ” had passed all stability tests for the IEC 61215 and IEC 61730 standards by third-party testing organization VDE of Germany, becoming the first in the world to receive certification. In addition, UtmoLight announced in February 2024 that its commercial-sized perovskite solar cell module with an area of 0.72 m² had passed the IEC 61215/61730 certification test and obtained the first and only TÜV certification.



The practical application of tandem technology using perovskite solar cells and c-Si solar cells is also actively being carried out around the world. Oxford PV (UK) started small scale production of tandem cells using M6 (166 mm wafers) in their German factory and plan to start commercial scale production in 2024. In June 2024, the company announced that it had recorded a conversion efficiency of 26.9% for a PV module with an area of approximately 1.6 m² (1 m x 1.7 m) (measured by FhG-ISE CalLab). The construction of a pilot line is advanced by Hanwha Solutions (Hanwha Q CELLS) (South Korea). The company established a pilot production line in Talheim, (Germany) in November 2022 through a consortium with the Helmholtz Center for Materials and Energy Research Berlin (HZB) and others. In May 2023, the company announced that it would invest \$100 million for the construction of a new pilot production line for tandem solar cell modules at its Jincheon plant in South Korea. The plan is to start commercial production by the end of 2026. Tongwei Group (China) plans to establish a photovoltaic industry technology research and development center in Sichuan Province, China, including a 100 MW/year perovskite/HJT tandem solar cell research and development line, and reportedly received approval from the Sichuan Provincial Development and Reform Commission in January 2024. The company announced that it had recorded a conversion efficiency of 31.68% for its perovskite/HJT tandem solar cell in November 2023. Other major Chinese companies are also working on research and development of perovskite/c-Si tandem technology, although they have not announced plans to build pilot lines. In June 2024, LONGi Green Energy Technology (China) announced that it had recorded a conversion efficiency of 34.6% for its small area perovskite/c-Si tandem solar cell (measured by EU ESTI). The company also announced that it achieved a 30.1% of conversion efficiency using M6 standard wafers (measured by FhG-ISE). Jinko Solar is also working on research and development of tandem solar cells. In May 2024, the company announced that a perovskite/c-Si tandem solar cell with an n-type TOPCon structure recorded a conversion efficiency of 33.24% (measured by SIMIT, China).

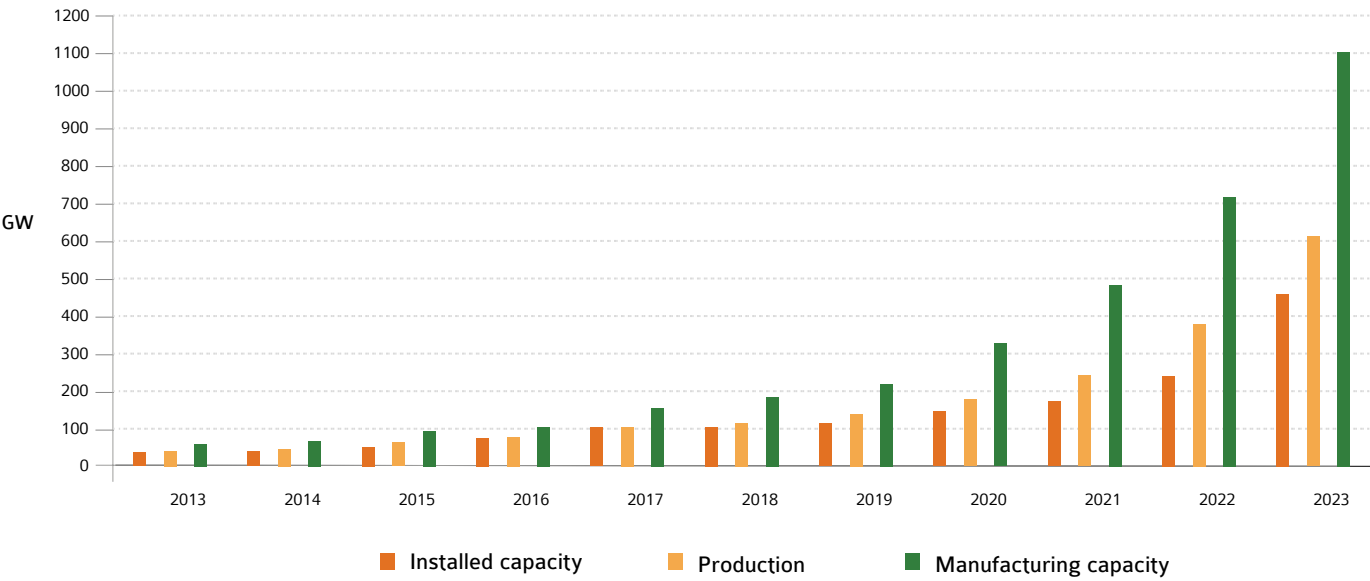
As mentioned above, research, development and demonstration of next-generation solar cells based on perovskite solar cells has been active, but neither single-junction nor tandem technologies have yet reached full-scale commercial production. It is necessary to improve conversion efficiency (to fill the gap between small-area cells and large-area modules), verify service life and stability, and establish a manufacturing process. In addition to overcoming these technical challenges, it is also necessary to identify target markets that take advantage of the characteristics of the technology, establish standardization (performance measurement and reliability evaluation), and product design as a system (construction method, etc.), as well as ensure environmental safety and respond to recycling efforts. For tandem types, durability and reliability must be verified before they can be introduced into the market of existing c-Si solar cells.

PRODUCTION CAPACITY EVOLUTION

Figure 4.7 and Table 4.3 show the evolution of global annual PV installed capacity, PV module production amount and PV module production capacity. In 2023, production capacity increased to 1 103 GW/year from 717 GW/year in 2022. The production capacity figures include the capacities of aging facilities and idle facilities that are not competitive anymore; hence the real effective production capacity is assumed to be on the level of about 700 GW/year to 800 GW/year in 2023. The speed of capacity enhancement is faster than the global PV market growth development so that the gap between demand and supply was widened in 2023 and PV module prices dropped. If the gap is not reduced and the inventory is not cleared current price levels are expected to remain.

THE UPSTREAM PV SECTOR / CONTINUED

FIGURE 4.7: YEARLY PV INSTALLATION, PV PRODUCTION AND PRODUCTION CAPACITY 2013-2023 (GW)



SOURCE IEA PVPS, RTS CORPORATION

**TABLE 4.3:** EVOLUTION OF ACTUAL MODULE PRODUCTION AND PRODUCTION CAPACITIES (MW)

YEAR	ACTUAL PRODUCTION			PRODUCTION CAPACITIES			UTILIZATION
	IEA PVPS	OTHER	TOTAL	IEA PVPS	OTHER	TOTAL	
	COUNTRIES	COUNTRIES		COUNTRIES	COUNTRIES		
1993	52		52	80		80	65%
1994	0		0	0		0	0%
1995	56		56	100		100	56%
1996	0		0	0		0	0%
1997	100		100	200		200	50%
1998	126		126	250		250	50%
1999	169		169	350		350	48%
2000	238		238	400		400	60%
2001	319		319	525		525	61%
2002	482		482	750		750	64%
2003	667		667	950		950	70%
2004	1 160		1 160	1 600		1 600	73%
2005	1 532		1 532	2 500		2 500	61%
2006	2 068		2 068	2 900		2 900	71%
2007	3 778	200	3 978	7 200	500	7 700	52%
2008	6 600	450	7 050	11 700	1 000	12 700	56%
2009	10 511	750	11 261	18 300	2 000	20 300	55%
2010	19 700	1 700	21 400	31 500	3 300	34 800	61%
2011	34 000	2 600	36 600	48 000	4 000	52 000	70%
2012	33 787	2 700	36 487	53 000	5 000	58 000	63%
2013	37 399	2 470	39 868,5	55 394	5 100	60 494	66%
2014	43 799	2 166	45 964,9	61 993	5 266	67 259	68%
2015	58 304	4 360	62 664	87 574	6 100	93 674	67%
2016	73 864	4 196	78 060	97 960	6 900	104 860	74%
2017	97 942	7 200	105 142	144 643	10 250	154 893	68%
2018	106 270	9 703	115 973	165 939	17 905	183 844	63%
2019	123 124	17 173	140 297	190 657	28 530	219 187	64%
2020	156 430	23 044	179 474	289 581	37 095	326 676	56%
2021	213 032	29 346	242 378	410 500	71 500	482 727	50%
2022	329842	48758	378600	611124	105476	716600	53%
2023	550992	61208	612200	1030500	72500	1103000	56%

Note: Although China joined IEA PVPS in 2010, data on China's production volume and production capacities in 2006 onwards are included in the statistics.

SOURCE IEA PVPS, RTS CORPORATION

THE UPSTREAM PV SECTOR / CONTINUED

Balance of system (BOS) component manufacturers and suppliers represent an important part of the PV value chain and BOS components are accounting for an increasing portion of the system cost as the PV module price is falling. Accordingly, the production of BOS products has become an important sector of the overall PV industry.

INVERTER

With the growth of the PV market in China that achieved a minimum of 235.0 GW and probably up to 277.2 GW¹ of installed capacity in 2023, more than double from the previous year, Chinese inverter manufacturers had a dominant position in the global market. Their dominance was observed in both utility-scale and distributed PV markets. According to CPIA, the global shipment shares of China-made inverters in 2023 was 70%. Huawei and Sungrow, both Chinese companies have about 50% of the global shipment whilst the Top 10 manufactures had more than 80% of the global volumes shipped. However, regional difference remains - the Chinese market is dominated by Chinese companies, but in the PV market in the Americas and Europe, European manufactures have a more significant share.

Generally, inverters are categorized into three types: central inverters, string inverters, and micro-inverters called “MLPE, module level power electronics”. In 2023, the share of central inverters (1 MW to 5 MW) used for large-scale utility projects above 10 MW was about 20%, down from 33% in 2022. The market share of string inverters used in utility-scale and small to medium-scale commercial PV systems was around 79%. The share of MLPEs remains low at about 1%, mainly for residential and small-scale commercial applications.

Inverter prices were under pressure so efficiency improvements have been advanced, achieving 99% and with the adoption of SiC or GaN, a reduction of the weight and size is observed. It is reported that new products achieved more than a 40% and 30% reduction of the volume and size respectively. In terms of the voltage, 1 500V products had an almost 55% share in the global market. In 2023, for the first time, 2 000 V products were connected to the grid in the Mengjianwan project in China. With higher voltages, 2 000 V PV inverters reduce power losses in the system and improve efficiency, not just of the inverter but also the power efficiency of transformers and other equipment, while reducing the cost of materials.

New grid codes require the active contribution of PV inverters to ensure grid management and grid protection. Grid forming, which requires sophisticated control and interactive communications functions with digital technologies and responds to inertia force is now being developed. While requirements are extended for inverter functions, serious concerns around cybersecurity are emerging, and addressing cybersecurity is now one of the major issues for energy security. Recently, US NIST published the draft guideline “Cyber security for smart inverters: Guidelines for residential and light commercial energy systems”. Solar Power Europe and other trade associations are also working on these issues.

Along with the growth of the distributed market, hybrid products that support power storage for residential and commercial use are increasing, and applications for EVs and virtual power plants (VPPs) are expanding. For self-consumption, functions to optimize self-consumption have been equipped, supported by energy management solutions combining energy storage systems and EVs with smart monitors. Applications of AI and machine-learning for failure detection as well as optimization of electricity generation contributed to lowering operation and maintenance (O&M) costs.

MLPE is growing in specific markets. Microinverters and DC optimizers (working on a module level) have been mainly adopted in the USA residential market to respond to rapid shutdown requirement imposed by the National Electricity Code (NEC) but are also being pushed in other markets where shutdowns and AC wiring is considered desirable (public buildings in France, for example). MLPE can help achieve a higher output for PV systems affected by shading and a more efficient rapid shutdown can be made in case of fire. In 2023, it is reported that the MPLE market in Europe grew with the demand for decentralized PV.

Regarding the plans of establishing PV manufacturing bases in the USA as mentioned above, the production capacity of inverters in the USA is expected to increase. In August 2023, Siemens announced contract manufacturing in USA and Enphase extended contract manufacturing in several states. Under the IRA, incentives are also provided for inverter manufacturing, and the IRA’s incentives are expected to increase demand for USA-made inverters, as there is a 30% investment tax credit (ITC) plus a 10% deduction for projects using domestic products.

1. A range of values is provided to account for uncertainty in AC/DC conversion ratios, in particular with regards to new utility-scale capacity in China, where the minimal annual volume considers official China reporting and the maximal annual volume considers a further 42 GW that could have been installed considering the uncertainty surrounding official conversion ratios from AC to DC of Utility-scale systems.



TRACKERS

Among other BOS segments, the market of single axis trackers has been growing. The installed capacity of single axis trackers of 2023 was around 70 GW. The largest tracker market is the USA, where more than 90% of the utility-scale projects adopted single axis trackers. Aside from the USA, the market for PV trackers is expanding in the countries where utility-scale PV projects are growing, China, India, Australia, South America, South Africa and the Middle East. Unlike other segments, the USA has the biggest share in tracker shipments accounting for more than 50% of the global single axis tracker shipments, but single axis trackers are also produced in demand areas such as China and Europe. Some of the principal companies (NextTracker, ArrayTechnologies and GameChange Solar) announced production capacity enhancements in the USA thanks to the dynamic and growing utility-scale market under IRA tax credits with a bonus for local contents. Besides utility-scale applications, trackers used for agriPV projects are developed and commercialized with specific designs to share solar energy with crops.

Companies doing business in the downstream sector have various origins: subsidiaries of electric utilities, subsidiaries of PV module manufacturers, companies involved in the conventional energy or oil-related energy business. Major PV project developers are accelerating their overseas business deployment and are active in business deployment in the markets such as Africa, the Middle East, ASEAN region and South America. The number of project developers active in the international business is increasing. Traditional energy companies are active in this field and TotalEnergies, Enel, Light Source BP, EDF Renewables and Iberdrola, etc. are developing PV globally. Chinese energy giants are also active in this field - Top 5 energy companies such as State Power Investment (SPIC), CHN ENERGY Investment Group, China Huaneng Group, China Three Gorges Renewables, and China Datang invested in around 80 GW of PV projects in 2023 in total. It should be also noted that several vertically integrated companies are present in the downstream sector. These companies produce PV modules or polysilicon, develop PV projects and provide EPC and O&M services. c-Si PV module manufacturers such as JinkoSolar, Trina Solar (both China), Canadian Solar (Canada), and Hanwha Solutions (South Korea) are also active in the downstream sector. On the other hand, in India, project developers entered upstream sectors to take advantage of PLI incentives. For example, in addition to PV cell and module manufacturing capacity, the Adani Group has plans to establish vertically integrated production from polysilicon to PV modules. In April 2024, a wafer and ingot manufacturing plant in Gujarat started operation.

THE UPSTREAM PV SECTOR / CONTINUED ..

Solar power remains the lowest-cost option in most countries, making it increasingly attractive to investors. While lowering PV module prices in 2023 stimulated the PV market, project developers need to consider recent inflation, rising interest rates, and increasing labor costs. In this situation, developers with strong credit and capital will likely become more competitive as they will have an easier time raising funds. Across the world, government tenders will need to consider recent inflation, rising interest rates, and fluctuations in product prices and devise flexible methods to continue to attract investment.



five

SOCIETAL IMPLICATIONS OF PV AND ACCEPTANCE

PV is bringing profound and undoubtedly long-lasting changes to modern society – with positive impacts across economic sectors, societies and the environment, it is a fundamental pillar without which the energy transition could not go ahead. As with all radical transformations, the acceptability – or even, the desirability – of the change is linked to perception of the attractiveness or necessity of the change and understanding of the impacts it will bring. For photovoltaics, this covers subjects such as the number of jobs involved, the creation of new companies and the disappearance or transformation of others, the generation of new financial flows, the impact on the environment and local communities. Understanding these social aspects related to the development of PV is becoming essential for governments adding photovoltaics to increasingly ambitious clean energy targets.

In particular, as penetration rates increase and utility scale and building integrated systems become more visible in local communities some populations are becoming less accepting of PV, whilst in specific countries, organised resistance to PV, most commonly for ground mounted systems but sometimes for building applied systems, has become a reality despite a generally positive opinion from the general public.

This chapter aims at providing key elements and indicators that can be used to promote the desirability of PV as an element of the energy transition while highlighting essential aspects that remain as sensitive points.

ACCEPTANCE OF PV DEPLOYMENT

Acceptance can be defined as the willingness of stakeholders to approve, support, and engage themselves in the energy transition. This acceptance is fuelled by a positive perception of the changes and decreased by negative inputs. In the early days of the development of PV, up until the European boom of 2007-2009, PV benefitted from an overwhelmingly positive image; it was developed with small scale distributed systems on roofs and was not a significant generator of revenue or tension.

Drops in acceptability have tended to come after boom growth – for example, in Spain in 2007-2008 the local feed-in tariff was so popular that PV developed so fast that local authorities cut support mechanisms, in fear of economic and budget consequences for the country. Other countries that stepped into the FiT policies have also experienced major market development followed by a rapid halt. The EU was the epicentre of PV development until 2011-2012, and this happened across a number of other European countries like Spain, France, Belgium, Czech Republic, Greece, Bulgaria and Romania.

Opposition in these contexts most often came from sectors that perceived photovoltaics as a threat: traditional utilities and energy sector majors for example, were unable to accommodate photovoltaics in their traditional business plans and pushed poorly informed authorities to put the brakes on PV development. The PV community had built its reputation on its environmental advantages and had not yet had to work on a broad social acceptance and was unable to work with governments to create healthy dialogue.

ACCEPTANCE OF PV DEPLOYMENT / CONTINUED

The most recurring arguments motivating a lack of social acceptance for PV depend on the country and the market segments, however common themes can be found including:

- Unappreciation of the physical appearance of PV systems in natural or heritage landscapes.
- Unfavourable opinions on the financial flows generated by PV systems (seen as either “profiteering” by individuals or multinationals profiting from local resources without contributing to local economies).
- Fear that toxic and/or rare materials are used in the manufacturing process with no possibility of recycling and/or that such materials may leach into the environment over time.
- Worries that PV will supplant crops and pose a risk to nutrition and food sovereignty.
- Opposition to ground mounted systems on the grounds of impacts to biodiversity and local environments.
- Concerns that developers will fail to take community sentiments or needs into account during system design
- Fear around quality and reliability issues (fire and electrical risks, resistance to storms...)

Where PV penetration rates are low, these issues seem to remain in the background, and only become prominent as market penetration increases. For those countries/regions where they are present, the PV sector, independent or government agencies have organised communication and educational campaigns, made fact-checking tools available to debunk the more aggressive false claims against PV and created educational resources targeted to the general and specific publics. Communication campaigns tend to be targeted towards the general public, working to demonstrate best practices, illustrate community investment and governance, or even educate the public about their ability to participate and engage in planning and development processes.

SOCIO-POLITICAL AND COMMUNITY ACCEPTANCE

In many countries there is a gap between national socio-political acceptance and community acceptance. These are associated with relatively different concerns and should therefore be addressed separately.

National socio-political acceptance refers to the acceptance of a technology by politics, policy makers, key stakeholders and the public. It englobes how legal and regulatory frameworks are adapted to include PV, and takes into consideration subjects as diverse as who carries the financial burden for support mechanisms

and grid capacity, jobs, industry development and local content. Socio-political acceptance is often lagging behind community acceptance in the early stages of PV development in a country.

Governments have demonstrated maturing levels of national socio-political acceptance through adapting planning legislation and procedures, reserving research budgets for community engagement or technology advancement, unrolling support for local jobs through deployment rules and local content. Examples Türkiye a hold on PV development until it was coupled with local value creation. In France, indirect local content requirements have been used in tenders for years, and local BIPV products were eligible for a FiT bonus over 2022 and 2023 – and in 2024 the two gigafactory projects for local manufacturing were declared Projects of National Interest. In the USA, recent funding programs improving community engagement in utility PV siting plans (R-STEP) and budgets specifically for research on community acceptance and opposition to solar (SEEDS) were implemented. In Japan, the 2023 meeting of the Agency for Natural Resources and Energy (ANRE) working group has developed recommendations for facilitating harmonious siting practices for local communities as well as provisions for suspending FiT and FiP for operators that did not respect all laws and regulations, working to reassure the general public of the accountability of generators.

Community acceptance is related to the acceptance by local stakeholders. It includes concerns over distributional justice (costs and benefits), procedural justice, and trust; NIMBYism (Not In My Backyard) sometimes occurs. It covers consideration of economic aspects: grid costs, Renewable Energy System (RES) fees, unequal access to PV, concentration of revenues between a limited number of big companies, social aspects (environmental, aesthetical impact), and specific opposition (e.g., farmers, hunters, lobbyists, etc). Many countries with well-developed utility scale markets are increasingly experiencing local and/or organised resistance to utility scale systems, with multiple factors cited as reasons for opposition, from the sealing of permeable lands to conflict of usage with agriculture or biodiversity reservoirs (Austria, France, Spain, Portugal, Sweden) – or even on the incompatibility of PV with the local cultural heritage (Italy). This opposition can be aimed at planning permits, but also become manifest in local planning regulations that restrict the siting of utility scale PV (USA, France, Italy, Netherlands).

Mandatory profit sharing of added value with local residents (France) is one type of action that can be taken to improve community acceptance, as is giving neighbours and community members the ability to participate in siting and planning procedures and consultations.



Challenges related to the acceptance of PV, even if they are directly influenced by the political, economic, geographical, social context in which PV installations are being deployed, are fairly similar across different regions and countries. This calls for a higher collaboration between countries on this topic based on the sharing of experience and exchange of good practices.

COMMUNITY ENGAGEMENT

Stakeholder engagement is an important way to improve acceptance and accelerate deployment. Stakeholders run across the value chain, from research down to permitting, construction and use – with stakeholder involvement important in some key areas such as permitting, grid connection and investment. Tools to increase involvement include public consultations in permitting procedures, self-consumption (encouraging all citizens to become generators), energy communities and collective ownership, open participation in the elaboration of climate and energy transition policies and targets or the provision of guidance and educational tools to end users.

The ability of local residents and communities to be involved in the process of zoning for solar, siting solar projects and the permitting procedure is now widely recognised as a lever for increasing community acceptance of large-scale projects. Accordingly, many governments have ensured that public consultations of local residents and communities are an integral part of the permitting procedures for PV projects, including in Austria, Japan, USA, France, UK. In Japan, from 2024 access to FiT and FiP for systems over a specific size or requiring forestry management authorisations is subject to a minimum number of public briefing sessions, and can include mandatory transparency on business plans. In the USA the government R-STEP (the Renewable Energy Siting through Technical Engagement and Planning) program will support the creation and/or expansion of state-based programs that improve renewable energy planning and siting processes for local communities. In France, public consultations are mandatory for ground mounted systems over 1MW. In Europe, the European Union adopted a series of new and updated guidance documents in May 2024 aimed at improving and streamlining permitting procedures to accelerate the deployment of renewables. These guidelines, recognising the importance of public participation for greater acceptability, recommend that “the needs and perspectives of citizens, local authorities and societal stakeholders should be taken into account at all stages of renewable energy projects from policy development to spatial planning and project development, deployment and operation.”¹

Citizen participation in energy communities or citizen investment and governance of solar projects is an important lever for both empowering residents and citizens to initiate projects but also maintain economic flows to local communities. In Austria, trials

have been undertaken to involve citizens in energy planning and investment. In France, the government co-finances the promotion of citizen investment in renewables. Citizen investment is present in many countries such as Austria, Germany, France, Denmark, USA, Australia... where citizens can either participate in controlling the project or receive financial participations.

Information and guidance web platforms and brochures can be used to facilitate citizen engagement in self-consumption, guiding through administrative procedures or demystifying the technology and providing buyers guide. In Australia the Solar Consumer Guide is a government website to guide through the process from installing to monitoring after installation; in the USA the Solar Energy Technologies Office of the Federal Energy Department hosts the Homeowner’s Guide to Going Solar website with the same goals; in France Photovoltaïque.info is a government financed website providing guidance from planning to operations for both end-users and professional installers; Singapore has a similar system, the government entity “Energy Market Authority” created a website that provides guidance on the installation steps, payment schemes and selling excess electricity. The installation steps section contains a list of the qualified persons (architects or professional engineers) with the permissions to install a PV system.

More generally, involvement can be seen under various angles:

- Individual participation as the beneficiary of PV electricity: Prosumers are consumers producing part or all of their electricity with PV while maintaining grid connection. Countries with prosumers policies, especially self-consumption ones are described in chapter 3. Energy access in emerging countries has shown for a long time that the implication of the populations significantly increases the adoption of decentralized energy sources.
- Individual participation as group actions for the development and use of PV electricity: Energy communities, and the specific case of solar communities are involving communities in producing and managing energy, allowing a higher involvement of stakeholders.
- Collectives and groups participating in the development of PV: Companies and utilities involved in the PV business are known to become advocates of the energy transition, as are local authorities that adopt PV as a tool in their climate change mitigation strategies.

1. Commission Recommendation (EU) 2024/1343 of 13 May 2024 on speeding up permitting procedures for renewable energy and related infrastructure projects

CLIMATE CHANGE MITIGATION

The paragraphs below highlight some key factual elements that can be used to improve the perception of PV in general, on economic, social and environmental aspects.

Climate change has become one of the key challenges that our societies has to overcome and PV is one of the primary solutions for reducing greenhouse gas emissions in the energy sector.

Global energy related CO₂eq emissions increased to 37 400 Mt² in 2023, up 1.6% on 2022, (compared to 0.9% from 2021 to 2022). Increasing the share of PV in the grid's electricity generation mix can significantly reduce the emissions from power generation. The global average carbon intensity of electricity was around 490 g CO₂/kWh³ in 2023 (slightly up on 2022) whereas for 1 kWh produced by PV the CO₂ emitted, taken on a life cycle basis, can be as low as 15 g depending on technology and irradiation conditions.

For example, the IEA PVPS Task 12 report from 2024 indicates that for the analysed configurations, in the case of the PV plant installed in the south of Italy (with an annual irradiation of about 1,820 kWh/m²/y), the estimated greenhouse gas emissions are 17.1 g CO₂ eq./kWh if the PV plant is equipped with mono-axial solar trackers and 20.7 g CO₂ eq./kWh if the modules are at fixed angle (34°).

The total CO₂ emissions that are avoided by PV on a yearly basis can be calculated considering the amounts of electricity that can be produced annually by the cumulated PV capacities installed at the end of 2023 and considering that these amounts replace equal amounts of electricity that would be generated by the respective grid mixes of the different countries where these PV capacities are installed.

The annually produced PV electricity is calculated based on country-specific yields depending on the average yields of PV installations and irradiation conditions in each country. With increasingly significant PV penetration rates in a large number of countries, and an overwhelming dominance of China manufactured modules, some changes have been made to the methodology used to determine avoided CO₂ in this report, compared to previous years reports.

Because of the higher penetration rates, this report now considers that PV production replaces electricity with a CO₂ content of the countries average grid mix (in the past, the CO₂ content of marginal production has been considered). This has significantly reduced the "avoided" CO₂, as PV now replaces average electricity, not just the most polluting easily dispatched electricity. Additionally, the inherent emissions of PV installed from 2022 has been uniformly considered as monocrystalline China-manufactured modules, independently of the country of installation⁴.

CO₂ avoided

923 MT CO₂eq

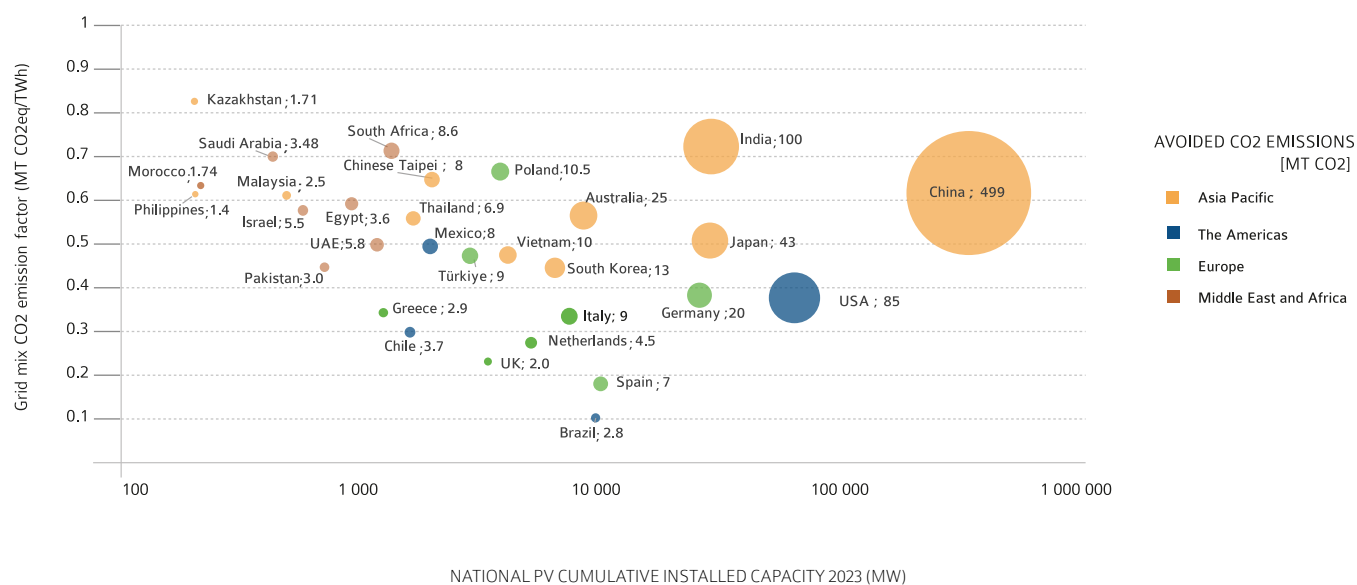
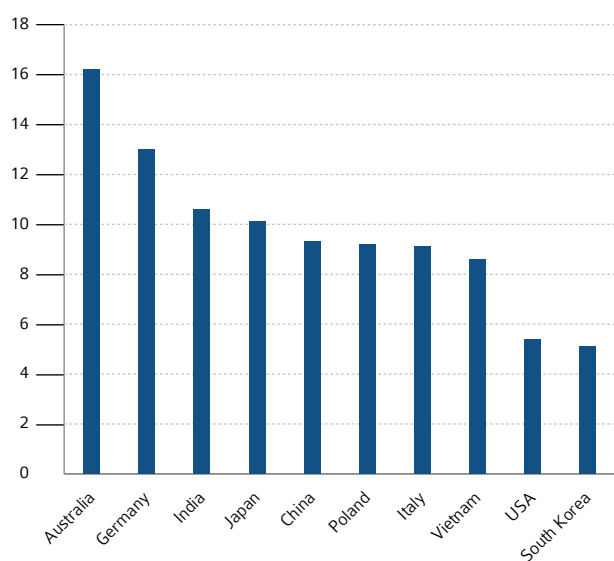
Using this methodology, calculations show that the PV installed capacity today avoids up to 923 million tonnes of CO₂eq annually. Thus, it avoids approximately 2.5% of the energy sector emissions. This is essentially due to the fact that PV is being massively installed in countries having highly carbon intensive grid mixes, such as China, India, Australia, Germany and Poland.

Figure 5.1 gives a view of the avoided CO₂ emissions in the first 30 countries in ranking of avoided CO₂ emissions, and which represent in total over 98% of global avoided emissions. This figure displaying the countries as a function of their installed PV capacities and grid mix carbon intensities clearly shows their differential contribution to the global avoided emissions and the high impact of their respective grid mix compositions. The more CO₂ the power mix in a country emits, the more positively PV installations will contribute to avoiding emissions.

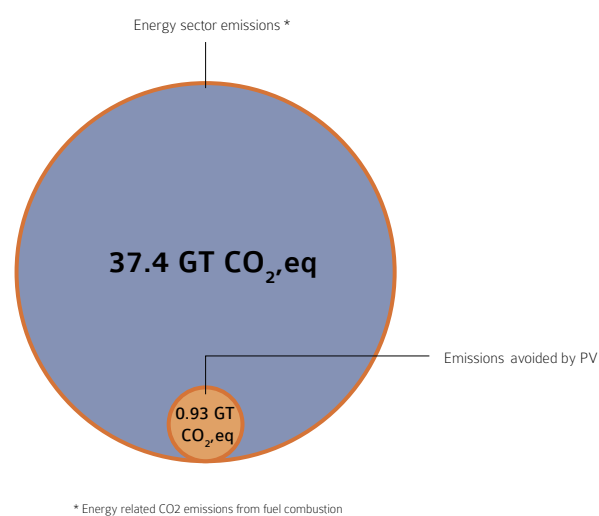
2. <https://www.iea.org/reports/co2-emissions-in-2023>

3. <https://ember-climate.org/insights/research/global-electricity-review-2024/>

4. Had previous years methodology been maintained, avoided CO₂ would have been reported as 1 927 MT CO₂eq., however with this updated methodology it is now reported as 8923 MT CO₂eq

**FIGURE 5.1: CO₂ EMISSIONS AVOIDED BY PV****FIGURE 5.2A: AVOIDED CO₂ EMISSIONS AS PERCENTAGE OF ELECTRICITY SECTOR TOTAL EMISSIONS**

SOURCE IEA PVPS & OTHERS

FIGURE 5.2B: AVOIDED CO₂ EMISSIONS AS PERCENTAGE OF ENERGY SECTOR TOTAL EMISSIONS

SOURCE IEA PVPS & OTHERS

VALUE FOR THE ECONOMY

The turnover of the PV sector in 2023 amounted to around 400 billion USD. This number has been calculated based on the size of the PV market (annual installations and cumulative capacities) and the average price value for installation and Operation & Maintenance (O&M) specific to the different market segments and countries.

Given the variety of existing maintenance contracts and cost, the turnover specifically linked to O&M has not been considered in detail. However, the global turnover related to O&M was estimated at around 15 billion USD per year. This estimate can be considered as a lower range value, due to the assumptions made for its calculations. It does not take into account either the material cost of replacement and repowering, which is hardly visible, or the value of recycling. O&M costs have decreased over time and a part of PV systems are not maintained through regular contracts (especially residential roof-top systems, unless they are monitored).

According to our estimates, in parallel to the growth of the annual market, the global business value of PV installations has increased by around 62%. The global value for O&M increased at least 50%. Between 2022 and 2023 the global business value increased an already considerable 25% - but this year's increase is well above that. Part of the increase is due to the reduction in system installation costs experienced in most markets in 2023 (even if the drop in module prices wasn't passed on to all markets in 2023), and part of it is due to the sheer size of the market increase. As in previous years, it is still important to note that this value increased less than annual new installed volumes increased (90%). It is worth noting that the O&M value is bound to grow further, powered by increasing volumes of centralised systems, aging plants and repowering operations.

Turnover PV

400 Billion USD

O&M

15 Billion USD

Global business value

+ 62% in 2023

For the purposes of this report, the value of the PV sector for the economy has been assessed based on the volume in MW of installations rather than by evaluating all the contributions of the complete value chain. The assessment of the business value of the industry is in general more complex, due to the decentralized production and the existence of transnational companies. However, a specific approximation of the industrial business value of PV was performed for IEA PVPS major PV manufacturing countries and is presented in a specific section below.

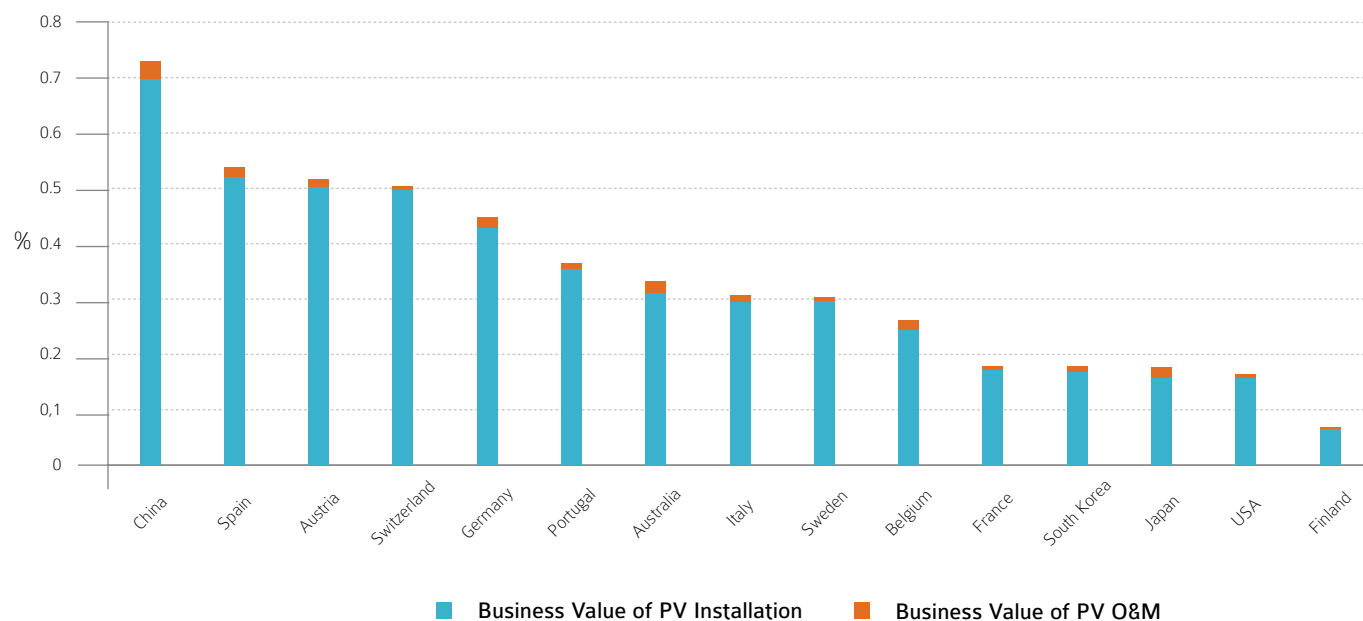
CONTRIBUTION TO THE GDP

Figure 5.3 shows the estimated business value of the PV sector in IEA PVPS reporting countries as compared to their national GDPs. These values were determined based on the internal PV markets in each country, as described above, and hence they do not take imports or exports into account. Some countries benefited from exports that increased the business value they obtained through the internal PV market while huge imports in other countries had the opposite effect. However, as already mentioned, the market is integrated to the point that it would be extremely complex to assess the contribution from each part of the PV value chain.

As shown by Figure 5.3, the business value of PV compared to GDP is growing as national market sizes increase – China, Spain and Austria all hit above 0.5% of GDP (this is +0.1% for China and over +0.25% for Austria), whilst a further six of the selected countries are above 0.3%. On a global scale, PV business value represents around 0.40% of the GDP, up from 0.25% in 2022, compared to around 2.96% for energy investments.

This level of investment is very close to global investments in all other generating technologies – but there is still scope for more, recognising that in the same year, over 1 000 billion USD were invested in fossil fuels.

In the IEA PVPS countries, the top 3 countries by volume of investment (China, USA, Germany) each roughly doubled their investment level, as did Italy.

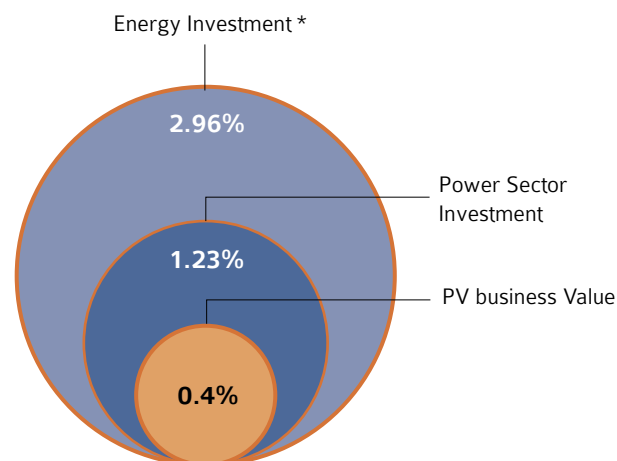

FIGURE 5.3: BUSINESS VALUE OF THE PV MARKET IN 2023 COMPARED TO GDP IN% IN 2023


SOURCE IEA PVPS & OTHERS

TABLE 5.1: BUSINESS VALUE: TOP 10 COUNTRIES IN ABSOLUTE VALUE

RANK	COUNTRY	BILLION US\$
1	CHINA	130
2	USA	44
3	GERMANY	20
4	SPAIN	8.5
5	JAPAN	7.5
6	ITALY	6.9
7	AUSTRALIA	5.9
8	FRANCE	5.4
9	SWITZERLAND	4.5
10	SOUTH KOREA	3.1

SOURCE IEA PVPS & OTHERS

FIGURE 5.4: CONTRIBUTION TO GLOBAL GDP OF PV BUSINESS VALUE AND ENERGY SECTOR INVESTMENTS


*Investment in the power sector, fuel supply and end-use & efficiency

SOURCE IEA PVPS & OTHERS

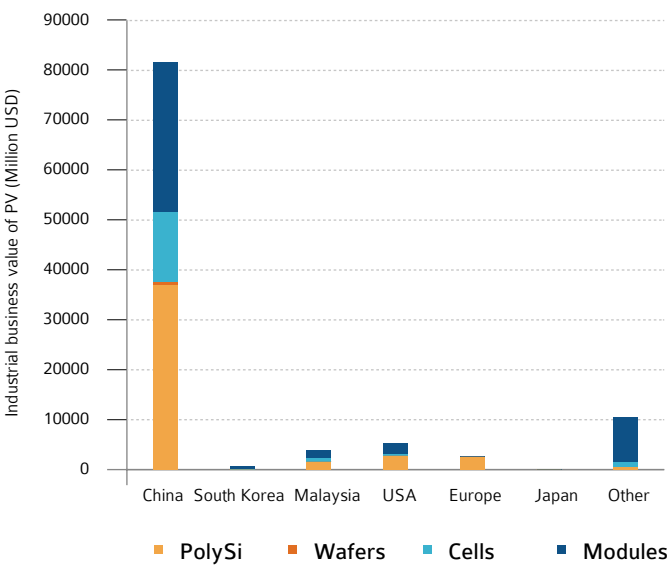
VALUE FOR THE ECONOMY/ CONTINUED

Even though assessing the detailed contributions of the different parts of the whole PV value chain is hardly possible in this report due to the level of integration of the market, an approximate evaluation of the industrial business value of PV has been performed and the results detailed for IEA PVPS major PV manufacturing countries.

The evaluation was made based on the production volumes and manufacturing shares of countries for polysilicon, wafers, cells and modules, including thin film technologies, as detailed in Chapter 4, as well as on an average estimated price for each of these four segments. The prices taken into account are based on average prices reported by member countries. We consider that equipment and materials are included in this computed value. BoS, including inverters are not considered here.

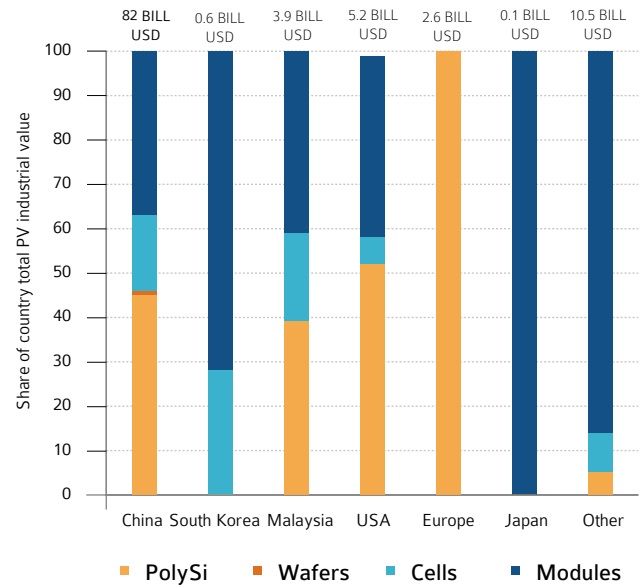
The estimated global industrial value of PV established itself at around 104.7 billion USD in 2023, up from 93.6 billion USD in 2022 (+12%). This relatively low growth is due to the significant price drop in average module and system costs in China (down to 0.14 USD/W or just 61% of the 0.23 USD/W cost in 2022). Figure 5.5A, 5.5 B and 5.5C show for IEA PVPS major PV manufacturing countries the estimated contribution of each step of the value chain in the PV industrial value for each country in absolute and relative terms as well as the comparison of this value to their GDP.

FIGURE 5.5A: ABSOLUTE PV INDUSTRIAL BUSINESS VALUE IN 2023



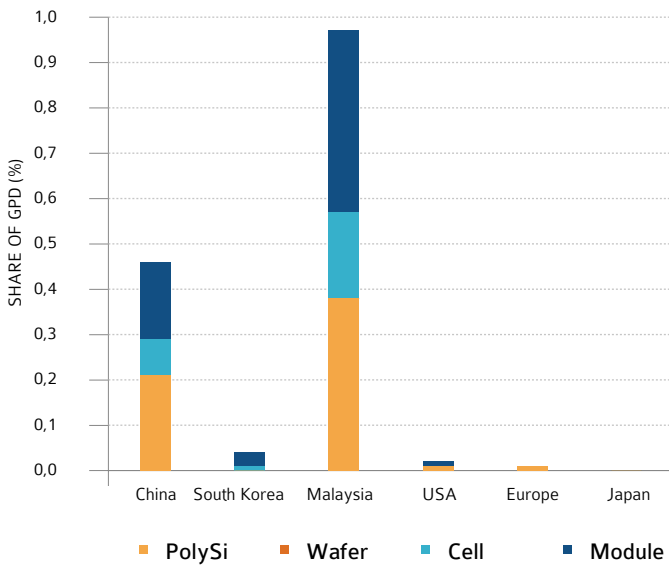
SOURCE IEA PVPS & OTHERS

FIGURE 5.5B: PV INDUSTRIAL BUSINESS VALUE ALONG THE VALUE CHAIN IN 2023



SOURCE IEA PVPS & OTHERS

FIGURE 5.5C: PV INDUSTRIAL BUSINESS VALUE AS SHARE OF GDP IN 2023



SOURCE IEA PVPS & OTHERS



China is by far the predominant manufacturing country in all steps of the PV value chain, with an approximate share of 0.46% of its GDP (up just 0.06% since 2022) represented by the PV Industry (polysilicon, wafers, cells and modules). As in previous years, despite much lower production volumes, the PV industry in Malaysia represents a significantly higher share of the country's GDP compared to China, stable at between 0.9% and 1%. As the many projects in the USA come online, the share in the USA is expected to become visible.

For the BoS, the industry is significantly more distributed, and production occurs in many countries. It is not counted as such here, with many local manufacturers and suppliers servicing the PV industry present across the world in cabling, supports and electrical protections; an analysis would make sense to grasp the extent of the PV industry impact on the countries' economic landscape but is not within the scope of this report.

SOCIAL IMPACTS

EMPLOYMENT IN PV

Figure 5.6 gives an overview of the total direct full-time equivalent jobs in IEA PVPS countries and India. Reported numbers have been established based on the IEA PVPS National Survey Reports and additional sources such as the IRENA jobs database. It should be noted that these numbers are strongly dependent on the assumptions and field of activities considered in the upstream and downstream sectors and represent an estimate in the best case and will diverge with those reported in other sources for this reason.

The methodology used takes data provided by reporting countries on the upstream (industrial) and downstream (distributed and utility scale PV installation and O&M) job numbers, then extrapolates to other markets depending on their respective market specifics. A distinction is made between countries in developed economies having a costly, low intensity work market and the emerging economies with an affordable work force, as well as economies somewhere between these two, where labour costs are mid-range. Manufacturing numbers are based on industry reports and additional sources and split according to the same methodology. Installation numbers are always an approximation.

This report estimates that the PV sector employed in the order of 7.2 million people globally at the end of 2023. An estimated 1.9 million were employed in the upstream part, including materials and equipment, while 5.3 million were active in the downstream part, including O&M.

PV sector employed an estimated
**7.2 million people in
2023**

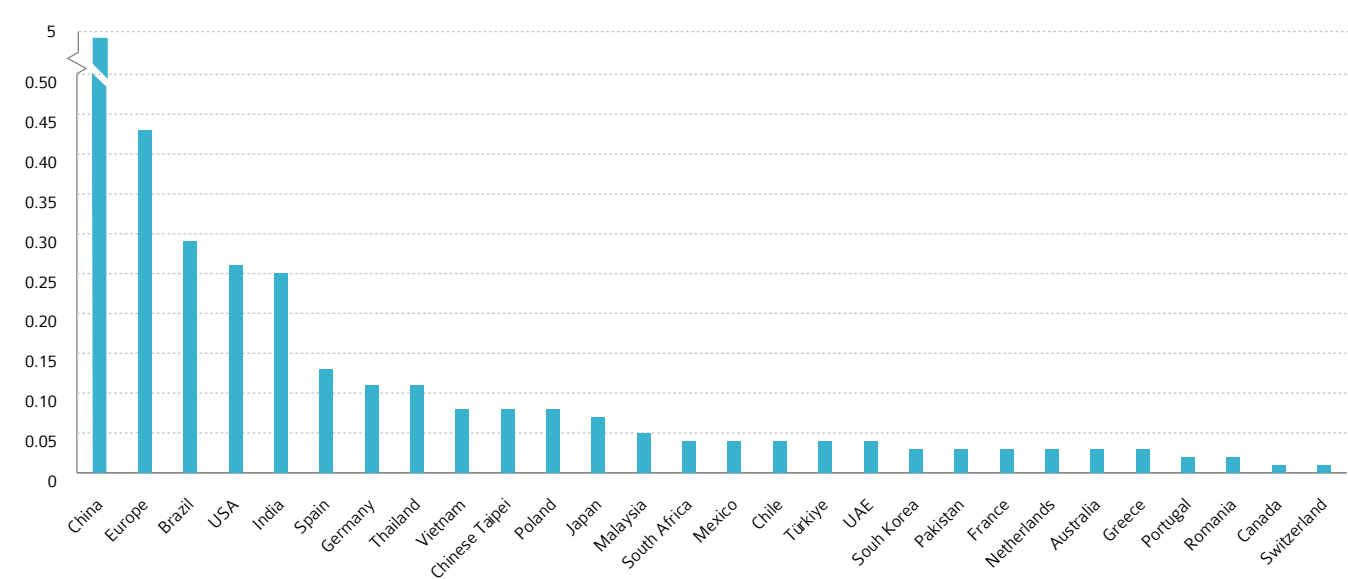
As the leading producer of PV products and the world's largest installation market by a long margin, China is markedly leading PV employment with around 5.1 million jobs in 2023, which corresponds to a significantly higher number of jobs than anywhere else. Lower by one order of magnitude, Brazil, the USA, and India follow, with between 250 000 and 300 000 FTE (Full Time Equivalent). There are an estimated 500 000 FTE in the European Union. Other countries with large workforces included Malaysia, Thailand, although estimations are difficult. Japan has a steady PV market and maintains approximately 70 000 jobs.

Generally, in good correlation with the market evolutions, PV employment expanded where the market developed. As the scale of manufacturing increases, so too does automation, reducing the job intensity per GW in the manufacturing sector. On the other end of the spectrum, residential (distributed) PV deployment has the highest job intensity along the value chain – although it varies significantly in intensity/MW between developing countries and developed countries.

When specifically focusing on development and installation activities, more labour intensive than manufacturing, on average the job intensity ranges from 3 FTE/MW for utility scale PV in developed countries with established markets, to up to 20 FTE/MW for distributed systems in developing countries.

SOCIAL IMPACTS / CONTINUED

FIGURE 5.6: GLOBAL EMPLOYMENT IN PV PER COUNTRY



SOURCE IEA PVPS & OTHERS

O&M generates many manual jobs while the entire PV value chain creates good quality jobs, from research centres to manufacturing.

With an estimated total of 7.2 million jobs in the solar PV sector worldwide in 2023, PV employs well above one third of the total renewable energy workforce and remains number one in the employment ranking of the global renewable energy sector, as new solar capacity additions largely outnumber other renewables sources, this is not expected to diminish.

LOCAL MANUFACTURING

The emergence of PV as a mainstream technology woke up appetites for local manufacturing and job creation at all levels of the value chain. Looking at IEA PVPS member countries only a few countries have pushed through different schemes for local manufacturing in recent years, namely Canada, Türkiye and the USA whilst others have at least initiated studies and advanced projects (France). Elsewhere, some countries have succeeded in bringing manufacturers to produce PV components in their country, for example in Malaysia, Vietnam, and Thailand. South Africa initiated local measures for partnerships with Chinese manufactures but have not yet been able to follow through. Other countries, such as Chile and Saudia Arabia seem to be thinking about it. With the disruptions in the PV value chain caused by the pandemic and the increased cost of shipment, the question of

local manufacturing gained traction in 2021 and remained in 2022 despite shipping costs returning to pre-pandemic levels early in the year. Countries such as India and the USA, to name a few, are pushing hard to develop a local industry and increase their partial independence. However, over 2023, increasing volumes of new manufacturing capacity came online in China, and the market cost of modules plummeted due to significant oversupply. As the low module costs were sustained through the end of 2023 and into 2024, different planned manufacturing facilities in Europe were cancelled (Germany), as investors called on EU countries to be more reactive faced with the Chinese oversupply and the attractive incentives offered in the USA. New manufacturing capacities require skilled workforces – and it is increasingly becoming a potential barrier; for example, Europe has identified the need to train at least 100 000 new people over 3 years for the PV industry.

IMPACT ON ELECTRICITY BILLS

The impact of developing PV on electricity bills works through three separate mechanisms; the cost of support mechanisms; the cost of electricity on wholesale markets and the cost of grid access fees.

Whilst in the past these three elements were seen as burdens, the increasing competitiveness of photovoltaics and the fluctuations of wholesale electricity costs over the past 3 years can allow one to consider them as opportunities:



Firstly, PV is reimbursing its support mechanisms: the impact of the increase in gas prices resulting from sanctions due to the war in Ukraine over 2022 and 2023 has resulted in PV (and other renewables) playing a much greater role in security of supply than intended so far. With spot market prices skyrocketing across Europe due to record high gas process, PV suddenly become not only competitive but desirable from an economical point of view. Those countries running support mechanism on Contracts for Difference even generated positive cash flow for governments, as renewable generators sold high on the market but reimbursed the state to reach agreed on contract levels for each MWh generated. In the UK, France, solar CfD contracts funnelled back millions of dollars to governments in 2022 (when market costs were exceptionally high) and in 2023 – despite market caps and reforms and lower market costs) in France, support mechanisms for solar raised 724 million euros (761 million USD) in 2022 and a further 81.3 million euros (88.9 million USD) in 2023 for the government and allowed the government to reduce taxes on electricity to contain prices for consumers. In Austria, fees to finance support mechanisms collected in electricity bills were cancelled in 2022. In the ACT in Australia, CfD contracts are anticipated to limited consumer electricity price to below inflation raises, under 5%, compared to a national average of over 20%.

Secondly, PV electricity generation is being sold on the markets in quantities sufficient to reduce market costs, especially in countries where peak consumption is concordant with solar generation. Numerous studies in various countries have shown that PV reduces wholesale market prices for electricity at the time of production – negative prices illustrate this and are most often seen when high solar (or wind) generation occurs at low load times. With the exception of Spain and Portugal, most of the Western European electricity markets saw dozens to hundreds of hours of negative prices in 2023, and are expected to see more in 2024 – with similar events in the USA, (South) Australia and China. The savings for electricity consumers and the society, in general, is difficult to compute but most studies conclude on significant savings and additionally, cost decrease in the distribution grid up to a certain penetration of PV. In the USA state of Texas's grid, it is projected that renewables will save \$6 billion USD in net system operating costs in 2024¹.

Thirdly, solar, generally combined with batteries, has demonstrated its ability to provide key network stability services cheaper than fossil fuel plants can – for example, in the nuclear-reliant France, delayed maintenance of the nuclear portfolio in 2022 led the transmission grid manager to call on renewables to step up in supplying services necessary to the grid, in particular voltage stability – whilst coal, heavy oil and additional gas facilities could have supplied these services, renewables were by far the most economical, and climate friendly, solution. In Australia, big batteries supply of frequency control ancillary services passed 50%

in 2023, fast making traditional coal or gas services redundant, whilst distributed batteries are reducing service loads on turbine generation. In the USA, solar power plants across several different states and extreme weather events have been key generators during crises.

PV FOR SOCIAL POLICIES

Besides its direct value for the economy and the jobs that it creates, both making contribution to the prosperity of the countries in which it is being installed and manufactured, PV entails additional positive implications on the social level if leveraged with appropriate policies.

PV can be a competitive alternative to increase energy access in remote rural areas not connected to power grids. Improved energy access can benefit rural business performance, free up workers' time, provide more studying hours for children, improve health through cleaner cooking, and create or enhance jobs as a result. Electrification is a key factor to reduce poverty and increase education, with a direct impact on women's and children's life standards in many regions in the world. In that respect, PV would deserve a significant attention for electrification.

In an increasing number of countries with stable electricity networks and close to total electrification, programs are or have been established to assist low-income families to install grid connected solar, either through means tested rebates, loans or gifts from state agencies or private organisations including in the USA at the national, state, and local levels, as well as Australia, UK. The development of energy communities is also being used as a tool to provide cheaper solar electricity to in need consumers in some countries such as Italy, Portugal. With the combination of high and volatile electricity prices experienced across many countries in the past 3 years, self-consumption of increasingly cheap solar is more and more often seen as the best solution to maintaining electricity bill affordability.

Increasingly cheap solar and self-consumption regulations offer opportunities for social welfare programs aimed at alleviating energy poverty. The energy crisis of 2022 increased the competitiveness of PV to the extent that it could reduce the electricity bill of families, municipalities and companies, not only through direct self-consumption but also through delocalized (or virtual) self-consumption.

Some more specific examples of energy poverty programs involving solar include:

In Malaysia, rural electrification is still a priority of the government, with a projected 100% electrification rate by 2025. Rural electrification is done together with utilities as a form of public-private partnership. In remote Sarawak, the Sarawak Alternative Rural Electrification Scheme (SARES) has electrified almost 5 000 households in 192 villages since its launch in 2016 and has received

1. https://poweralliance.org/wp-content/uploads/2023/04/NERA-Astrape-White-Paper-20230405_Final.pdf

SOCIAL IMPACTS / CONTINUED

regional recognition in 2019. Solar PV and hybrid systems are often used in this scheme, as well as micro hydro-technologies.

In South Korea, with the financial aid from Seoul Metropolitan government, a non-profit organization, Energy Peace Foundation, and Solar Terrace company installed 30 kW of mini-PV systems for 100 energy-vulnerable households (300 W/household). This type of mini-PV installations is becoming popular in South Korea to reduce the electricity bill burden during the summer.

In Italy, the government has established a fund to finance self-consumption systems of between 2 kWp and 6 kWp for households in economic difficulty across 2024 and 2025; a municipality in Sardinia collaborated with an energy services operator in 2017 where the municipality purchased PV systems then sold on loan to families in energy poverty conditions, to make them benefit from PV self-consumption and thus reduce their energy bills. The revenues of the net-billing feed a public fund, in order to finance the maintenance of the plants or possibly the purchase of other plants for other families. The scheme has been replicated in other municipalities, and an energy poverty observatory has been set up.

In Australia, a number of measures were announced by State Governments in 2020 and have been maintained in 2023 going from interest free loans or even complete subsidies (Solar for Low Income Households for systems with an installed capacity up to 3 kW). Additional measures tackling rural electrification include a budget to support feasibility studies looking at microgrid technologies to replace, upgrade or supplement existing electricity supply and to finance the deployment of PV to reduce the use of diesel.

In France, rural electrification is addressed in overseas territories and isolated alpine areas through budgets available for off-grid electricity production, electric vehicle charging points, or grid-connection financing, whilst the widespread availability of micro-PV kits (1 or 2 modules) is leading to small investments by many low income families. Social housing organisations have been granted simplified conditions for managing collective self-consumption systems on social housing as an indirect incentive to allow low income families to benefit from self-consumption.

In the USA, the Inflation Reduction Act (IRA) contained 145 million USD in grants, 18 billion USD in loans, and carve-outs within the low-income bonus to the Investment Tax Credit for Tribal solar deployment to address disparity of electricity access for indigenous communities. The IRA also contained significant incentives for rural electrification, as well as measures targeted towards energy access for low-income communities, historically marginalized communities, and communities with higher rates of unemployment as a result of the energy transition.

In Brazil, the program “Luz para todos” was launched in November 2023, with the goal of improving rural electrification through renewable energies. In its new phase “Luz Para Todos Minha Casa Minha Vida”, the program aims to install 500 000 solar panels in low-income households across the country. The primary objective of this new phase is to lower electricity costs while increasing access to renewable energy.

In Chile, the Energy inclusion program founded in 2018, brings together international collaboration and public-private funds to create concrete actions to improve the electricity conditions of vulnerable households under innovative business models adapted to their actual conditions, strengthening local economic development and labor insertion. One example of these projects includes the financial support to build a solar refrigeration center for a fishing village.



PV END-OF-LIFE

The volume of PV modules reaching the end of their useful (first) lifetime is still marginal compared to the volumes of new PV modules deployed in the market. However, as the PV market develops fast and often faster than anticipated, the same trends are expected to be witnessed for end-of-life PV module streams. Forecasting precisely end-of-life PV module streams is a complex exercise for several reasons. PV modules may reach the end of their useful lifetime for different reasons – significant performance degradations, premature failures from production defects, damages from transportation and installation or premature dismantling related to insurance claims, repowering, or revamping. These modules then enter end of life streams anywhere from after just one year or up to thirty years or more. A large disparity in useful lifetime is observed between distributed and centralized applications, with shorter useful lifetimes observed in centralized systems, mostly driven by economic considerations, i.e. the hardware is uninstalled due to financial lifetimes and not technical failures. The market for second-life PV modules is a further source of uncertainty in end-of-life PV module streams forecasts.

Depending on the country and region, end-of-life PV modules may be treated under PV-specific regulations or under general waste and disposal-related regulations.

In the EU, end-of-life PV module streams are regulated by the WEEE (Waste Electrical and Electronic Equipment) Directive since 2012. The Directive is based on the extended producer responsibility principle that stipulates those producers (the term broadly refers to manufacturers, distributors, sellers and importers) placing PV modules on the EU market (regardless of where the PV modules were manufactured) are liable for the costs of PV waste collection, treatment and monitoring. Producers can choose to operate their own take-back and recycling scheme or join existing ones. The WEEE Directive sets collection, recovery and preparation for re-use and recycle minimum requirements, which are expressed in percentage by mass.

Recycling requirements are currently typically achieved through mechanical processes which rely on

1. removal of some components (e.g., frame, junction box, cables),
2. mechanical shredding,
3. sorting into different material categories, taking advantage of physical property differences (weight, conductivity, density, ...) amongst the recovered materials (plastics, glass, metals).

These mechanical recycling processes are usually performed by incumbent recycling actors (e.g., EEE recyclers, metal recyclers, glass recyclers) who leverage existing recycling facilities, equipment and expertise, eventually reaching recycle at a relatively low net cost while enabling WEEE-compliant recovery rates. Processes based on delamination (mechanical delamination (e.g., hot-knife) or thermal delamination (e.g., pyrolysis, incineration)) also exist and are implemented at a commercial level in some rare cases only. Combined with some subsequent chemical process, such recycling routes have the potential to recover materials with higher levels of purity (e.g., glass, silicon) or to recover high-value or critical materials (e.g., silver). However, they are associated with higher net costs and the WEEE Directive requirements are not stringent enough to provide a regulatory-push for such processes.

In other regions, country specific approaches have been taken. In Asia, in China, two demonstration lines for PV waste recycling were set up after a 2019-22 national R&D program focused on recycling crystalline silicon PV modules, and in April 2022 the PV Recycle Industry Development Center in Jiaxing, Zhejiang province was set up as a public institution affiliated with the Ministry of Industry and Information Technology. In Australia, in some states, PV modules are banned from landfill and must be treated in the electronic waste streams. Limited facilities exist to undertake recycling, however a Solar PV Stewardship pilot project was opened first for decentralized PV waste and then utility scale waste in 2024. In Japan, from July 2022, setting aside of future cost of EoL PV systems became compulsory for solar power generation facilities with more than 10kW installed capacity under the FIT program. Owners of PV systems who fail to make reserves for dismantling and removal of PV modules may be subject to revocation of FIT. The Organization for Cross-regional Coordination of Transmission Operator, OCTT, is responsible for managing the reserves. Part of the reserve is expected to cover the cost of recycling of PV modules. In September 2021, the Agency for Natural Resources and Energy under the Ministry of Economy, Trade and Industry published a guideline. In the USA by early 2024 there were nearly 30 recyclers listed in an Office of Energy Efficiency & Renewable Energy website, built as part of the National PV Recycling Program founded in 2016.

PV END-OF-LIFE / CONTINUED

Thin film cadmium-telluride panels, which represent a smaller part of the solar market, undergo a specific recycling process with a USA manufacturer running dedicated recycling facilities for thin film panels which recover the semiconductor material (cadmium and tellurium) in addition to glass and copper. The Inflation Reduction Act also contained limited incentives for the construction of recycling facilities for renewable energy technology, and several announcements of recycling facilities have been made since its passage in August 2022.



six

COMPETITIVENESS OF PV ELECTRICITY IN 2023

The rapid price decline that PV experienced in the last years has already opened possibilities to develop PV systems in many locations with limited or no financial incentives. However, the road to full competitiveness of PV systems with conventional electricity sources requires answering many questions and bringing innovative financial solutions, especially to emerging challenges.

This section aims at defining where PV stands regarding its own competitiveness, starting with a survey of module and system prices in several IEA PVPS reporting countries. Given the number of parameters involved in competitiveness simulations, this chapter will mostly highlight the comparative situation in key countries. Prices are often averaged and should always be looked at as segment related.

The question of competitiveness should always be contemplated in the context of a market environment created for conventional technologies and sometimes distorted by historical or existing incentives. The fast development of nuclear in some countries in the last 50 years is a perfect example of policy-driven investments, where governments imposed the way to go, rather than letting the market decide. The oil and gas markets are also perfect examples of policy-driven energies which are deemed too important not to be controlled. PV competitiveness should therefore be considered in this same respect, rather than the simple idea that it should be considered competitive without any regulatory or financial support. There are also further barriers, other than economic, for PV to become the obvious alternative to coal (rather than gas) for utilities. Currently, many already unprofitable coal power plants are still in operation because the regulatory and financial structure is not tailored for so many coal units to become stranded assets. In addition, the choice of alternatives to coal is frequently not motivated by pure economics but is biased towards an electricity

price and market design that favour gas-fuelled electricity. Since all sources of electricity have benefited at some point from such support, the question of the competitiveness of PV should be considered carefully. Hereunder, we will look at the key elements driving the competitiveness of PV solutions.

MODULE PRICES

The very first period of PV market development can be considered starting from the first prototypes to small-scale production leading to a total PV installed capacity of around 2 GW. During this first phase, price reductions corresponding to a learning rate of 20% were achieved: this allowed the total PV installed capacity to continue growing further. At that point, prices stabilized until the total capacity reached around 10 GW: this period is known as the time of low availability of polysilicon that maintained prices at a high level. Then, a third period started which is still the case today, beginning with the mass production of PV, especially in China. During this period ranging from 10 GW to current levels, significant economies of scale have led to a learning rate of 40% - the supply chain disruptions due to COVID having slowed it slightly. The present price drops due to overcapacity have compensated in part for the stable prices in the post-covid period. In a few years it will be possible to better evaluate the impact of the significant manufacturing capacity increases that occurred in 2023 and 2024.

MODULE PRICES / CONTINUED

FIGURE 6.1: PV MODULES SPOT PRICES LEARNING CURVE (1992-2023)

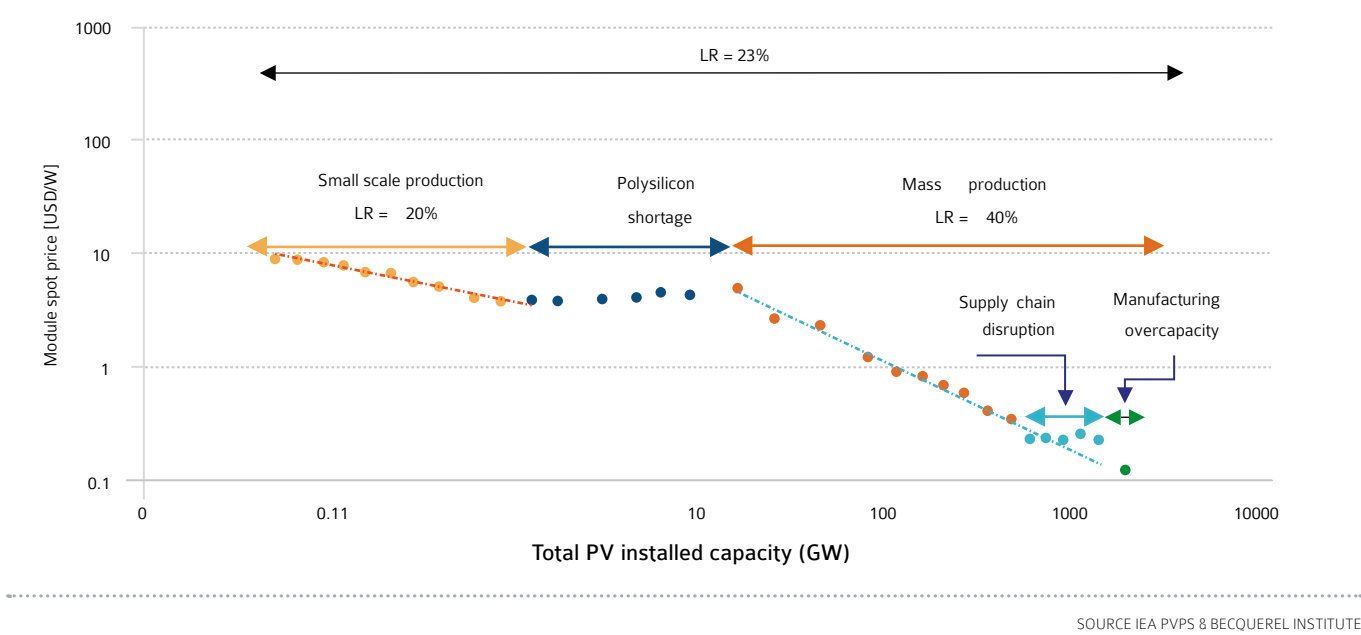
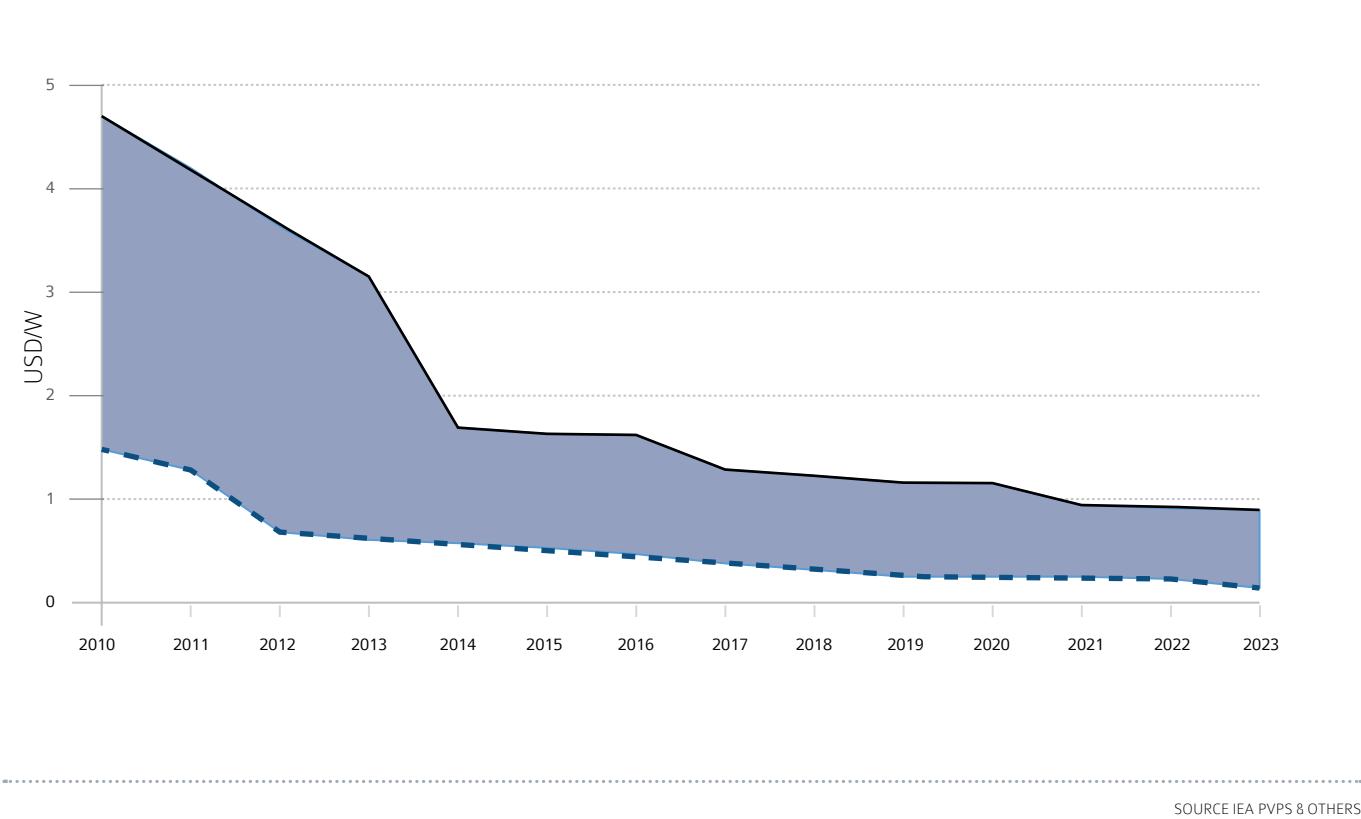


FIGURE 6.2: EVOLUTION OF PV MODULE PRICES RANGE IN USD/W





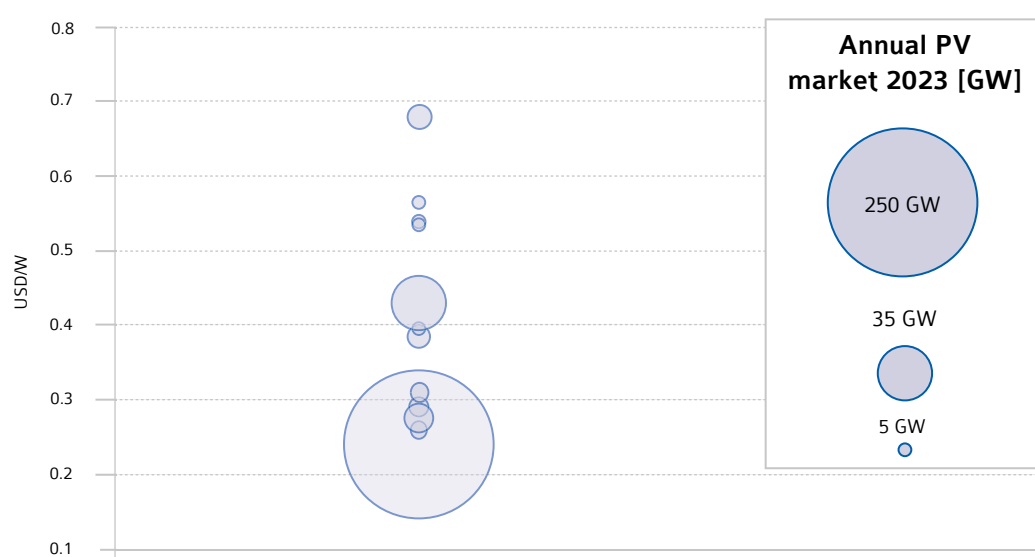
On average, the price of PV modules in 2023 (shown in Figure 6.3) accounted for approximately 40% and 50% of the lowest achievable prices that have been reported for utility scale systems. In 2023, the lowest price of modules in the reporting countries was around 0.14 USD/W, significantly below anything ever seen before. This price is a direct consequence of the large volumes of manufacturing that came online in 2023, well above the market's capacity to absorb. Low as it was, prices continued to drop into 2024 and went under 0.10 USD/W.

Looking back to understand past and present price trends, the Chinese decision to strictly limit PV subsidies in May 2018 led to a new imbalance between production and demand, with dozens of GW of new production capacities added in 2017 and 2018 in all segments of the value chain while the global PV market was stagnating. The price decrease that followed accelerated some project development and can be considered at least partially responsible for the market growth in 2020. The year 2021 had seen the rise of multiple raw material prices. In particular, PV polysilicon average spot prices rose significantly during the year, up from around 10 USD/kg in early 2021. Other key raw materials such as PV glass, copper or aluminium maintained their high prices reached at the end of 2020 and the whole PV value chain

was subject to increases in transportation costs. Costs and hence prices remained high through 2022, and manufacturers upgraded capacity through this time. By 2023 new manufacturing capacity was coming online as there was a polysilicon price drop that cascaded through to cells and modules, and with a surge in module availability, prices began to drop. By the end of 2023, and despite important efforts in China to stimulate the market enough to absorb the surge in manufacturing capacity, oversupply was impacting the module price for Chinese-made modules. The consequences of the policy decisions in the USA in years previous to the ramping up of manufacturing capacity meant the bulk of Chinese modules - that were not installed domestically - were exported to Europe. By late December 2023, mainstream modules had reached costs that analysts considered under manufacturing cost. Looking in depth at the revenues of some manufacturers among the most competitive, it appears that average sales are above these low prices. It can also be assumed that such prices are obtained with new production lines in which production costs are significantly lower than previously existing ones.

Higher module prices are still observed, specifically in the residential segment in countries with historically high prices such as Japan or France.

FIGURE 6.3: INDICATIVE MODULE PRICES IN SELECTED REPORTING COUNTRIES



SOURCE IEA PVPS & OTHERS

SYSTEM PRICES

Reported prices for PV systems vary widely and depend on a variety of factors including system size, location, customer type, connection to an electricity grid, technical specifications, and the extent to which end-user prices reflect the real costs of all the components. For more detailed information, the reader is directed to each country's national survey report on the IEA PVPS website (www.iea-pvps.org).

Figure 6.4 shows the range of system prices in the global PV market in 2023. It shows that around 75% of the PV market consists of prices below 1 USD/W. Large distributed PV systems start at around 0.70 USD/W while utility-scale PV saw prices as low as 0.42 USD/W. BIPV can be seen as a series of segments where the prices can significantly diverge. Off-grid applications suffer from a similar situation, with totally different cases illustrated at different prices. In general, the price range decreased from the previous year for all applications.

On average, system prices for the lowest-priced off-grid applications are significantly higher than for the lowest-priced grid-connected applications. This is mainly attributable to the relatively higher transport costs to access the sites. Indeed, large-scale off-grid systems are often installed in places far from the grid but also far from major towns and highways. Higher prices asked for such installations also depend on higher costs for the transport of components, and technicians, without even mentioning the higher costs of maintenance. In 2023, the lowest system prices in the off-grid sector, irrespective of the type of application, typically ranged from about 2 USD/W to 6 USD/W but prices for some specific

applications can be higher. The large range of reported prices in Figure 6.5 is a function of the country and project-specific factors. The highest prices haven't been included in the figures given the very low level of installations: in general, off-grid prices have been averaged in the figures for readability reasons.

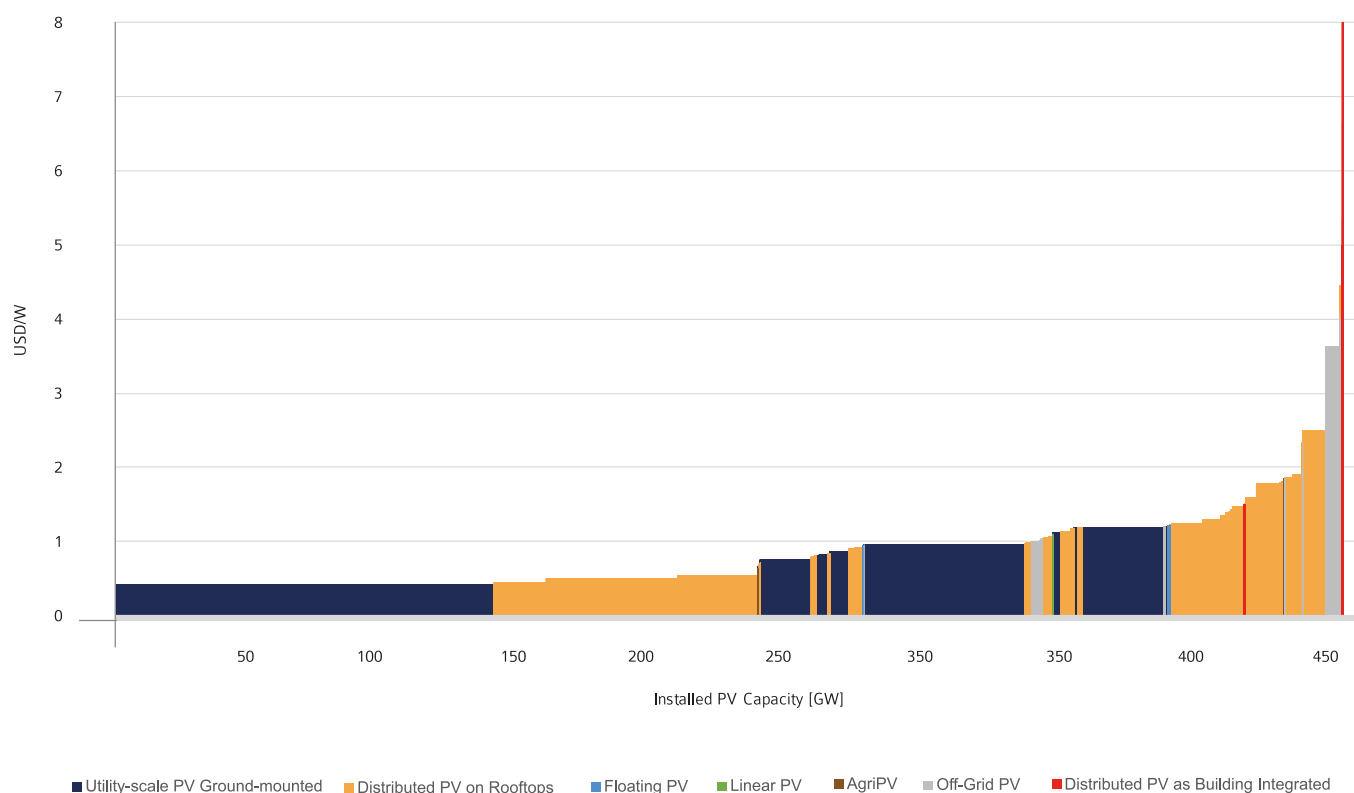
Four specific segments are developing that are likely to grow significantly in volume in countries with limited land. System prices in these segments can vary widely, both from local labour and material costs but also given the inherent constraints specific to the sites used.

- Floating PV – costs vary with anchoring for local weather, salinity, system size and grid connection.
- Linear PV systems along roads, trainline and canals – costs vary depending on size, electrical architecture and grid connection, and can be as low as ground mounted utility scale systems but tend to be higher.
- Parking canopies over commercial and industrial car parks – costs vary depending on the type of supports and system size, but are above ground mounted system costs.
- Agrivoltaics either in inter-row grazing or above crops – costs vary from as low as utility scale ground mounted systems when associated with grazing up to BIPV-level costs for specialised mobile systems above crops

More expensive grid-connected system prices are often associated with roof-integrated slates, tiles, building integrated designs or single bespoke projects: BIPV systems in general are considered more expensive when using dedicated components, however the drop in module prices is likely to impact some, but not all, BIPV products.



FIGURE 6.4: 2023 PV MARKET COSTS RANGES



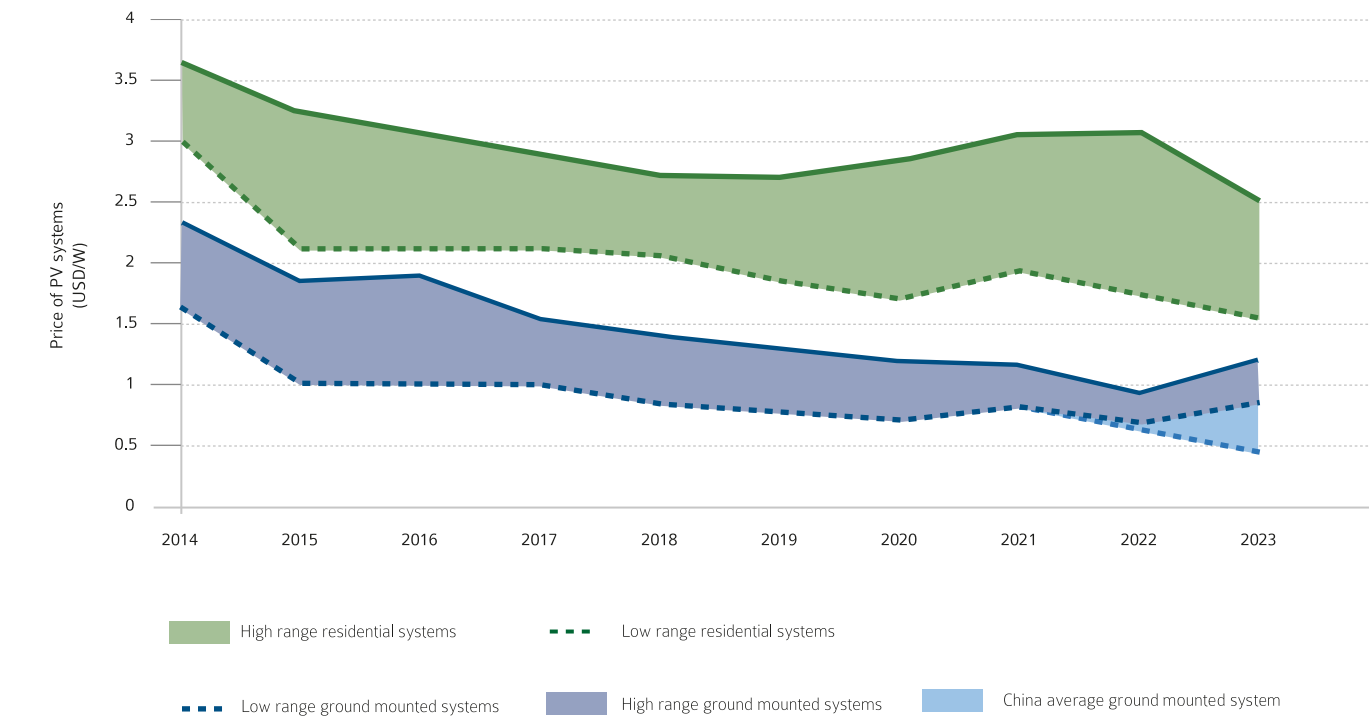
Note: Utility and distributed PV are distributed according to volumes and costs. Floating, Linear, AgriPV and Offgrid PV are indicative average maximum / minimum price points; the real costs are spread between these two points with outliers.

SOURCE IEA PVPS & OTHERS

In 2023 residential PV systems price typically ranged from 0.78 USD/W to 2.6 USD/W (with costs in China below this and costs above this in Switzerland) while utility-scale PV systems prices typically ranged from 0.42 USD/W to 1.2 USD/W in 2023 according to the data collected – a wider range than in 2022, but expected given the module price drop between the beginning and end of the year. These typical price ranges give an overview of the market, but they don't consider the full range of prices practiced

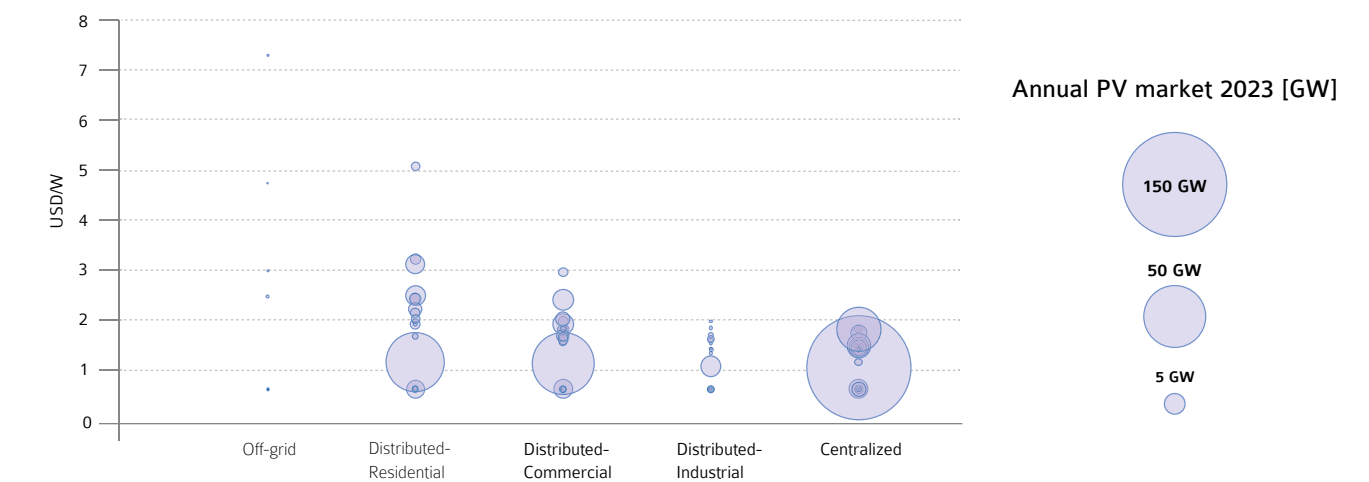
across the world – residential systems have, in reality, a much wider range depending on local contexts, with prices below this range depending on labour and administrative costs and the size of systems, and prices going above this range for systems using BIPV products, high end inverters and monitoring or simply scarce labour. Utility scale systems can also go beyond this, particularly on the bottom range, when project developers work aggressively to secure land or for particularly large systems.

FIGURE 6.5: EVOLUTION OF RESIDENTIAL AND GROUND MOUNTED SYSTEMS PRICE RANGE 2013 - 2023 (USD/W)



SOURCE IEA PVPS & OTHERS

FIGURE 6.6: INDICATIVE INSTALLED SYSTEM PRICES IN SELECTED IEA PVPS REPORTING COUNTRIES IN 2023



SOURCE IEA PVPS & OTHERS



The lowest achievable installed price of grid-connected systems in 2023 also varied between countries as shown in Figure 6.6. The average price of these systems is tied to the segment. Large grid-connected installations can have either lower system prices depending on the economies of scale achieved, or higher system prices where the nature of the building integration and installation, degree of innovation, learning costs in project management and the price of custom-made modules may be considered as quite significant factors. In summary, system prices for utility-scale PV mostly decreased by the end of 2023, following the trends of module prices and the balance of the system, while soft costs and margins remained stable, and interest rates rose. System prices below 0.6 USD/W for large-scale PV systems were common in very competitive tenders and in China. The question of the lowest CAPEX is not always representative of the lowest LCOE (Levelized Cost of Electricity): the case of utility-scale PV with trackers illustrates this, with additional CAPEX translating into a generally lower LCOE.

Additional information about the systems and prices reported for most countries can be found in the various National Survey Reports.

COST OF PV ELECTRICITY

In order to compete in the electricity sector, PV technologies need to provide electricity at a cost equal to or below the cost of other technologies – unless other criteria are a determining factor. Obviously, power generation technologies are providing electricity at different costs, depending on their nature, the cost of fuel, the cost of maintenance and the number of operating hours during which they are delivering electricity.

The competitiveness of PV can be defined simply as the moment when, in a given situation, PV can produce electricity at a cheaper price than other sources of electricity that could have delivered electricity at the same time. Therefore, the competitiveness of a PV system is linked to the location, the technology, the cost of capital, and the cost of the PV system itself, which highly depends on the nature of the installation and its size. However, it will also depend on the environment in which the system will operate. Off-grid applications in competition with diesel-based generation will not be competitive at the same moment as a large utility-scale PV installation competing with the wholesale prices on electricity markets. In sum, the competitiveness of PV is connected to the type of PV system and its environment.

GRID PARITY

Grid Parity (or Socket Parity) refers to the moment when PV can produce electricity (the Levelized Cost of Electricity or LCOE) at a price below the price of electricity bought from the grid. While this is valid for pure players (the so-called “grid price” refers to the price of electricity on the market), this is based on two assumptions for prosumers (producers who are also consumers of electricity):

- That PV electricity can be consumed locally (either in real-time or through some compensation scheme such as local or delocalized net metering);
- That all the components of the retail price of electricity can be compensated when it has been produced by PV and locally consumed.

Technical solutions can be used to increase the self-consumption level (demand-side management including EV charging or direct use to heat water with heat pumps, local electricity storage, reduction of the PV system size, delocalized self-consumption, energy communities, etc.).

If only a part of the electricity produced can be self-consumed, even after actioning levers for increased self-consumption, as mentioned above, then the remaining part must be injected into the grid and should generate revenues of the same order as, at least, any centralized production of electricity. Today this is often guaranteed for small size installations by the possibility of receiving a FiT (or similar) for the injected electricity through net billing schemes. However, when grids are under strain, when local or regional generation exceeds demand, it is possible that excess electricity either cannot be injected into the grid (i.e. must be curtailed) or will not be financially compensated – as may soon be the case in parts of Australia, Austria, USA, amongst other countries experiencing grid constraints in the residential and commercial sectors.

The price paid for electricity by consumers is composed in general of four main components:

- The procurement price of electricity, on electricity markets plus the margins of the reseller;
- Grid costs and fees, partially linked to the consumption, partially fixed; the key challenge is their future evolution;
- Taxes;
- Levies (used among other things to finance the incentives for some renewable sources, social programmes, solidarity between regions etc.);

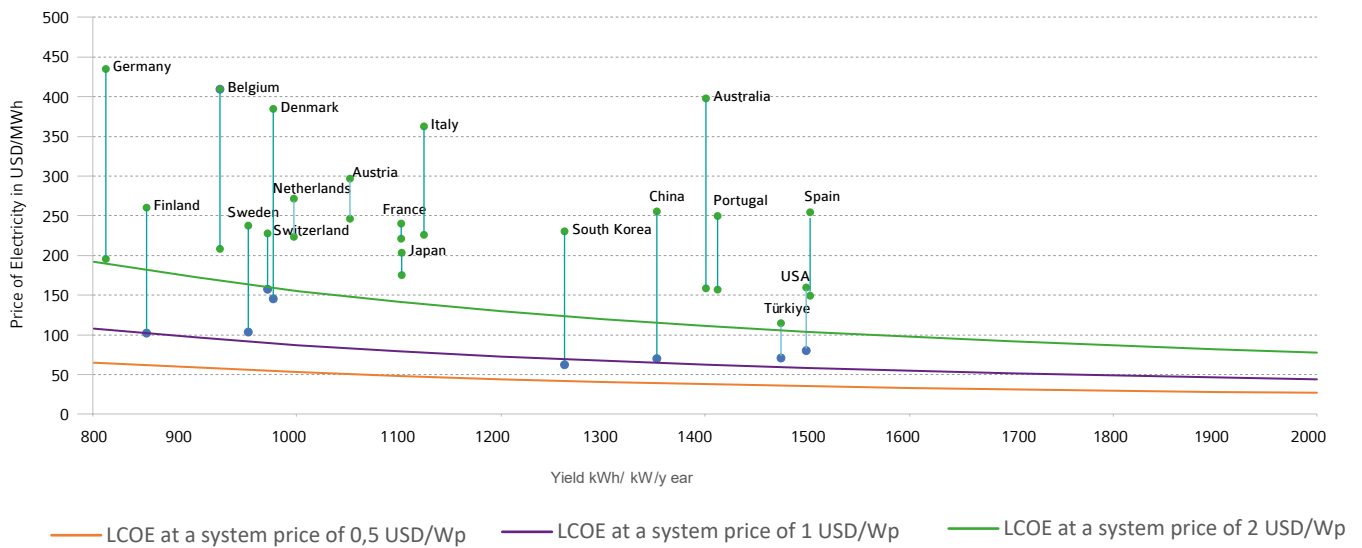
If the electricity procurement price can be compensated, the two other components require considering the system impact of such

COST OF PV ELECTRICITY / CONTINUED

measures: with tax loss on one side and the lack of financing of distribution and transmission grids on the other. While the debate on taxes can be simple, since PV installations are generating taxes as well, the one on grid financing is more complex. Even if self-consumed electricity could be fully compensated, alternative ways to finance the grid should be considered given the loss of revenues for grid operators - or a better understanding of PV positive impacts on the grid should be achieved.

Any support or compensation measures for the excess electricity after self-consumption should take into account these last elements and the question of equitable and ethical sharing of the cost burden for grids and supports schemes should become an increasingly important subject as PV penetration rates increase.

FIGURE 6.7: LCOE OF PV ELECTRICITY AS A FUNCTION OF SOLAR IRRADIANCE & RETAIL PRICES IN KEY MARKETS*



*NOTE: THE COUNTRY YIELD (SOLAR IRRADIANCE) HERE SHOWN MUST BE CONSIDERED AS AN AVERAGE SOURCE IEA PVPS & OTHERS

THE LOWEST ELECTRICITY PRICES (RESPECTIVELY THE HIGHEST) DISPLAYED PER COUNTRY SHOULD BE SEEN AS AN AVERAGE VALUE FOR INDUSTRIAL CONSUMERS (RESPECTIVELY RESIDENTIAL CONSUMERS).

Figure 6.7 shows the range of retail prices in selected countries based on their average solar resource and the indicative PV electricity threshold for three different system prices (0.5, 1 and 2 USD/W, converted into LCOE). Green dots are cases where PV is competitive in most of the cases. Blue dots show where it really depends on the system prices and the retail prices of electricity.

The figure demonstrates how grid parity had been reached in most IEA PVPS countries through late 2022 and 2023, and how rising electricity costs and dropping PV costs are paving the way for more countries becoming competitive for PV. In this figure, the lower range of electricity prices is, with the exception of France, Türkiye for commercial and industrial consumer, whilst the higher is

for residential customers. In 2022, retail and commercial/industrial electricity prices rose quite steeply in most countries, increasing the competitiveness of PV – and whilst in many countries, particularly in the EU, commercial electricity prices dropped slowly through 2023 and again in 2024, the drop in module prices means that competitiveness is here to stay.

In some specific segments, determining the competitiveness of a PV system is not just dependent on the system or electricity costs, but also on other avoided costs, as for BIPV, for example. The specific case of BIPV consists, for new or renovated roofs, to assess the competitiveness for the BIPV solution minus the costs of the traditional roofing (or façade) elements. The rest of the



assessment is similar to any building under self-consumption using a standard BAPV solution. Of course, if the BIPV solution has to be installed on a building outside of any planned works, this doesn't apply. Metrics used for buildings can also be different, since the integration of PV components might be justified by non-economic factors or the perspective of an added value. For such reasons, BIPV competitiveness is in general assessed against the traditional building costs.

COMPETITIVENESS OF PV ELECTRICITY WITH WHOLESALE ELECTRICITY PRICES

In countries with an electricity market, wholesale electricity prices when PV produces are one benchmark of PV competitiveness. These prices depend on the market organisation and the technology mix used to generate electricity. In order to be competitive with these prices, PV electricity has to be generated at the lowest possible price. This is already achieved with large utility-scale PV installations that allow reaching the lowest system prices today especially when they come with low maintenance costs and a low cost of capital – although interest rates rose around the world in 2023, increasing the cost of capital. An increasing number of countries have had utility scale systems commissioned that sell directly on the market – including Croatia, Italy, Germany, Norway, Sweden, Portugal, Romania, Philippines, Spain, Australia and the USA; indeed, in these last three countries it is increasingly becoming a common option.

Energy-only markets are also likely to be soon complemented by grid services that could add additional revenues.

These types of business models, exposed to the market, remain riskier than conventional ones that guarantee prices paid to the producer over 15 years or more. The key risk associated with such business models lies in the evolution of wholesale market prices on the long term and the potential effects of what is known as price cannibalisation: large volumes of PV generated electricity reduce market prices during the midday peak when penetration becomes significant. However, with high penetration rates and the shift to electricity for transport and heating, it is unclear if price cannibalisation will be a real issue or not – either prices during PV production peaks will drop and impair the ability to remunerate investments, or low prices will attract additional electricity demand and will stabilise the market prices. At this point, both options remain possible. When a wholesale market doesn't exist as such, (in China for instance), the comparison point is the production cost of electricity from coal-fired power plants.

FUEL-PARITY AND OFF-GRID SYSTEMS

Off-grid systems including hybrid PV/diesel can be considered competitive when PV can provide electricity at a cheaper cost than the conventional generator. For some off-grid applications, the cost of the battery bank and the charge controller should be considered in the upfront and maintenance costs while a hybrid system will consider the cost of fuel saved by the PV system.

The point at which PV competitiveness will be reached for these hybrid systems takes into account fuel savings due to the reduction of operating hours of the generator. Fuel-parity refers to the moment in time when the installation of a PV system can be financed with fuel savings only. It is assumed that PV has reached fuel-parity, based on fuel prices, in numerous Sunbelt countries.

Other off-grid systems are often not replacing existing generation sources but providing electricity in places with no network and no or little use of diesel generators. They represent a completely new way to provide electricity to hundreds of millions of people all over the world.

PRODUCING COMPETITIVE GREEN HYDROGEN AND OTHER MOLECULES WITH PV

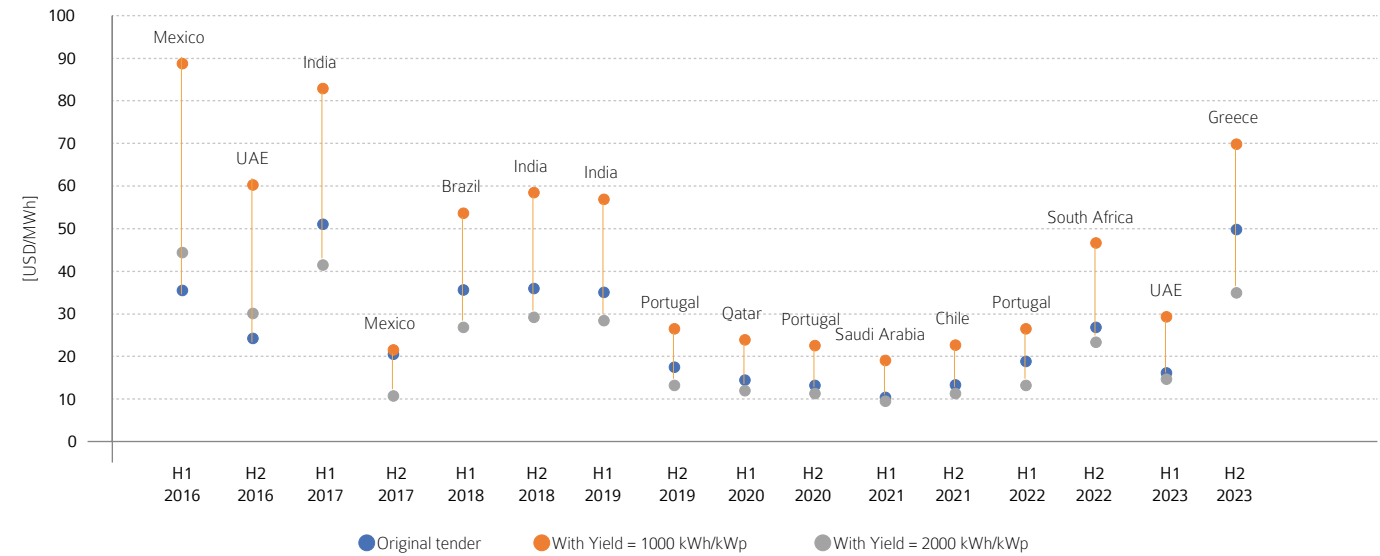
The declining cost of PV electricity opens the door for other applications such as green hydrogen and/or ammonia directly from PV (possibly in combination with wind). The cost of electrolyzers is also decreasing as commissioned volumes of electrolyzers increases; however, despite the increased gas prices after the war in Ukraine, competitiveness with "black" hydrogen is still only on the horizon and is not yet a reality. Different uses for hydrogen – industrial applications, transport, agriculture (through ammonia) – are expected to create a tremendous opportunity for PV to produce hydrogen without being connected to the grid.

Large projects have been announced around the world with several commissioned already including two systems in China with 150 MW / 300 MW of PV and a smaller project in Japan. The business model is being explored in a wide range of countries, with increasingly large multi-GW scale projects in preliminary phases across Australia, the USA, Brazil and Africa (Mauritania) and the Middle East (Oman, Egypt) – countries where high irradiance and low land costs are expected to allow for competitive conditions.

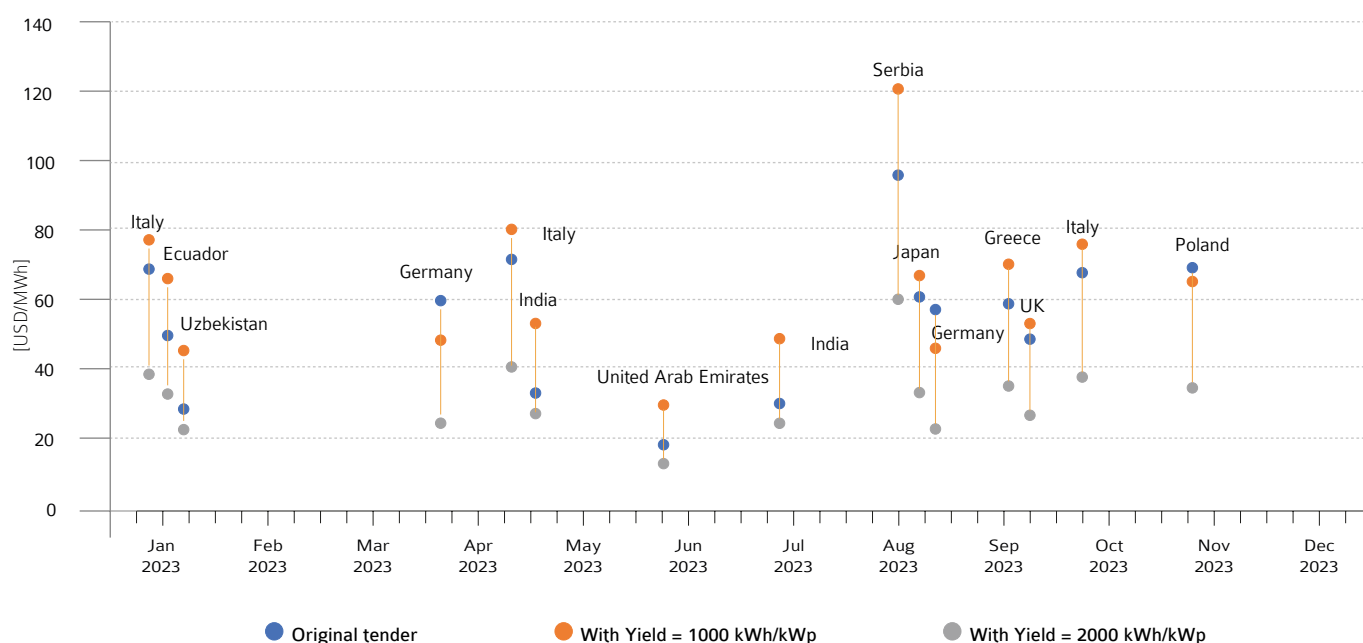
COST OF PV ELECTRICITY / CONTINUED

LOW TENDERS PRICES

FIGURE 6.8A: NORMALISED LCOE FOR SOLAR PV BASED ON LOWEST PPA PRICES 2016 - Q4 2023



SOURCE IEA PVPS & OTHERS


FIGURE 6.8B: NORMALISED LCOE FOR SOLAR PV BASED ON RECENT PPA PRICES 2023


SOURCE IEA PVPS & OTHERS

With many countries having adopted tenders as a way to allocate PPAs to PV projects, the value of these PPAs achieved record low levels in 2020 and some low prices in 2021 as well. In 2022 and 2023, because of the higher electricity market prices (Europe, USA, Australia) PPA's were negotiated at higher prices too. The high level of certainty when it comes to solar generation levels and operating costs has become attractive on markets where many industrial and commercial consumers were significantly impacted by the volatile electricity market prices of late 2022 and early 2023. It is increasingly worth paying a slight premium for the price stability. In parallel, solar is an increasingly attractive option for companies investing in renewable energy for renewable energy targets and corporate social responsibility goals such as the RE100 initiative.

Solar PPA prices seen in 2023 ranged from about 20 USD/MWh to 100 USD/MWh – a slightly wider range than in 2022, with a clear difference in regional evolution; they increased quarter on quarter through 2022 in Europe and the USA, and either stabilised or slightly decreased in 2023 (in Europe) whereas they continued to increase through 2023 in the USA. Whilst in the past most solar PPA's were for new and to be developed systems, in the coming years they will be for existing capacity reaching the end of current contracts under support mechanisms or initial PPA's.

COST OF PV ELECTRICITY / CONTINUED

TABLE 6.1: TOP 10 LOWEST WINNING BIDS IN PV TENDERS FOR UTILITY SCALE PV SYSTEM

REGION	COUNTRY/STATE	USD/MWH	YEAR
MIDDLE EAST	SAUDI ARABIA	10.40	2021
EUROPE	PORTUGAL	13.20	2020
LATIN AMERICA	CHILE	13.32	2021
MIDDLE EAST	UNITED ARAB EMIRATES	13.53	2020
MIDDLE EAST	QATAR	14.49	2020
MIDDLE EAST	SAUDI ARABIA	14.80	2021
EUROPE	SPAIN	14.98	2021
EUROPE	PORTUGAL	16.02	2019
MIDDLE EAST	UNITED ARAB EMIRATES	16.20	2023
LATIN AMERICA	BRAZIL	17.50	2019

SOURCE IEA PVPS & OTHERS

TABLE 6.2: LOWEST WINNING BIDS IN PV TENDERS FOR UTILITY SCALE PV SYSTEM PER REGION

REGION	COUNTRY/STATE	USD/MWH	YEAR
ASIA	UZBEKISTAN	17.9	2021
AFRICA	TUNISIA	24.4	2019
EUROPE	PORTUGAL	13.2	2020
LATIN AMERICA	CHILE	13.3	2021
MIDDLE EAST	SAUDI ARABIA	10.4	2021
NORTH AMERICA	MEXICO	20.6	2017

SOURCE IEA PVPS & OTHERS



seven

PV IN THE ENERGY SECTOR

PV ELECTRICITY PRODUCTION

TRACKING OF PV INSTALLED CAPACITY

Tracking PV installations in all the regions of the world can be challenging as many countries do not accurately keep track of the PV systems installed or do not make the data publicly available. Data published by IEA PVPS reports on new annual installed capacity and total cumulative installed capacity and is based on official data in reporting countries. For the purposes of this report, when individual countries do not publish any reporting data, estimated volumes have been made based on trade data (import volumes), and adjusted to take into account the delays between importing volumes and installation and commissioning.

Depending on reporting practices, cumulative capacity (the sum of new annual capacity) may outstrip operating capacity as systems are decommissioned. Repowered capacities not only replace some decommissioned capacity but also generally increase operational capacity, as the repowered capacity is higher than the initial plant capacity due to PV module efficiency improvements.

There is no standardised reporting on these subjects across IEA PVPS countries. Several countries already incorporate decommissioning of PV plants in their total capacity numbers by reducing the total cumulative number. Other countries report capacity in operation for that year, and do not include repowered volumes in new annual capacity or decommissioned volumes in operational capacity. Many countries do not track decommissioning or repowering with any consistency.

Repowering is still relatively unusual given the age of the oldest installations, but it is expected to increase in the near future – serial defects with backsheets manufactured in the period 2009 – 2011 is a good example, as the past 3 years have seen several hundred MW replaced. Module capacity that has been used to repower systems with defective or underperforming modules will appear in shipped volumes but not necessarily in new annual installations. Real decommissioning is expected to be rare, as land usage constraints and cheaper PV on buildings encourages repowering. Recycling numbers can provide a glimpse of what is happening with regards to repowering and decommissioning in countries where recycling schemes are active, however recycling volumes are underestimating decommissioning due to an active (and sometimes barely legal) second-hand market, especially towards Africa; also, reporting is often in tonnage and the availability of data must be improved before it can be used more generally.

IEA PVPS is following the dynamic evolution of decommissioning, repowering and recycling closely, however giving numbered estimates is not possible yet, so the expected impact on the installed capacity, market projections for repowering and the decline in PV performances due to ageing PV systems is not quantified.

ESTIMATING PV PRODUCTION

Estimating PV electricity production is easy to measure at a power plant but much more complicated to compile for an entire country. Not only the installed capacity must be accurately tracked, which requires an effective and consistent approach (especially for distributed, self-consumption and off-grid segments), but also, electricity production is impossible to accurately estimate from installed PV capacity for a given year, because estimations are based on the installed capacity without knowledge of when, during the year, that capacity was installed. A system installed at the end

PV ELECTRICITY PRODUCTION / CONTINUED

of the year will have produced only a small fraction of its theoretical annual electricity output.

Additionally, estimates are based on a theoretical annual production, and do not take into account different performance levels based on azimuth and inclination or even other factors such as system ventilation or shading. A system installed west facing will generate less over the year than one installed facing the equator. Performance losses due to aging of PV plants are not considered at this point. Some plants may have experienced production issues, due to technical problems or external constraints.

For these reasons, the electricity production from PV per country in this report is an estimate of what the minimal theoretical production should be the following year, when all the PV systems installed at the end of the year have generated electricity for one year.

To calculate this “next year’s minimal theoretical PV production”, an average solar yield in the country is used. Depending on the country, this value is either provided through National Survey Reports and is calculated as the total solar generation divided by total solar capacity, or is a representative value taken from geographical irradiance atlases – these yields are an approximation of the reality.

Nationally produced statistics on solar generation will in general reflect real production injected into the grid – consequently, in years with high new capacity additions, the theoretical value used here can be well above official statistics for the current year – and well below that of the next year. The real PV production in a country is increasingly difficult to assess even if tracked by transmission system operators (TSO’s), as more and more volumes of generation are self-consumed (i.e. not metered) and storage enters into consideration, increasing self-consumption levels or curtailing generation to fit storage. IEA PVPS advocates for governments and energy stakeholders, including grid operators to create accurate databases and precisely measure PV production.

PV PENETRATION

PV electricity penetration can be three different indicators – either the installed capacity per capita, as seen in Chapter 2; or, as used here, the share of electricity consumption (or demand) or electricity production supplied by PV generation. Here it is the ratio between PV electricity production in a country and the electricity demand in that country and is expressed as a percentage. It is based on the theoretical electricity production from PV per country, calculated as indicated in the section above. Electricity demand is obtained via publicly available databases and via the IEA PVPS experts; when possible, it has been adjusted to consider self-consumption volumes.

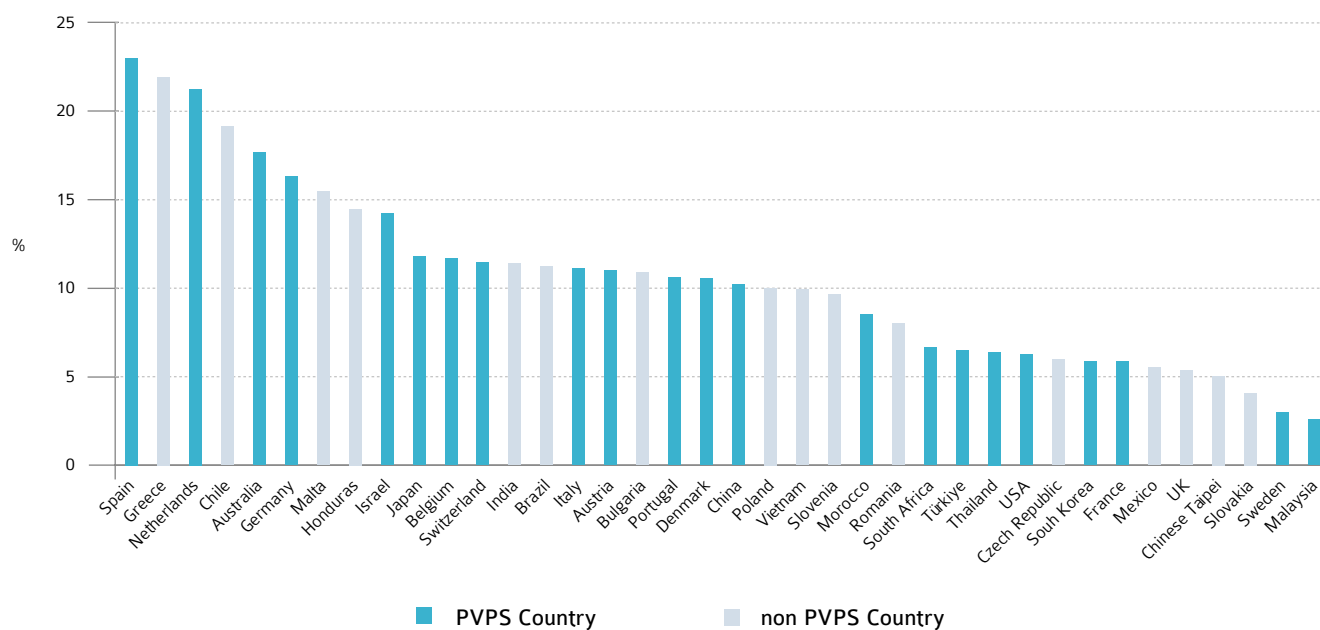
The PV penetration rates here are an estimate and are likely to differ from official PV production and penetration numbers in many countries for the reasons detailed above – they should be considered as indicative, providing a reliable estimation for comparison between countries and do not replace official data.

The PV penetration rate calculated based on electricity consumption will tend to be several points higher than when calculated based on electricity production, as losses from transport and transformation of centralised (fossil fuel) electricity generation have not yet been factored in.

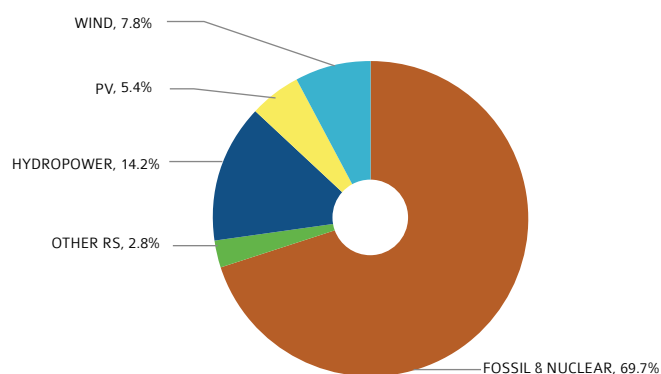
To illustrate this divergence, the national TSO reported on production data in France, and gave the PV penetration at 5% for 2023, just above last year’s IEA PVPS estimation of 4.9% but under estimations for this report at 5.8%. In Spain the TSO reported it as 14% for 2023, well below our estimated 19.7% last year and 22.3% this year. The difference from Spain is due to the fact that the TSO does not take into consideration the large volume of distributed PV in Spain – consequently, the divergence is expected.

Of the 41 IEA PVPS and non-IEA PVPS countries studied, all but one (Norway) produced at least 1% of their electricity demand from PV in 2023 – and 10 should be producing more than 10% next year.

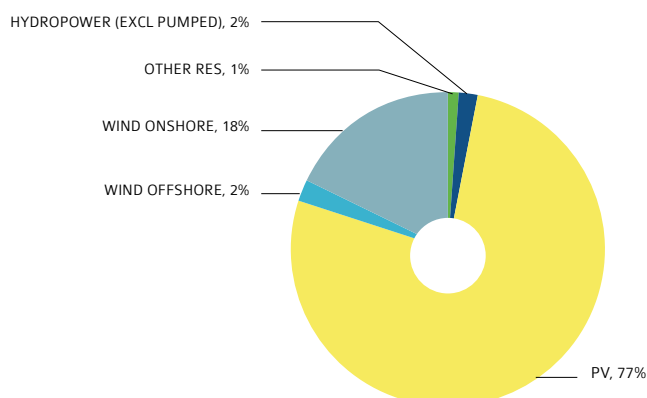
Concerning global PV penetration, with around 1 642.0 GW installed worldwide, PV could produce almost 2 136 TWh (see Table 7.1) of electricity on a yearly basis if an average yield of 1300kWh/kWp is considered. This represents around 8.3% of the global electricity demand covered by PV, up 2.1% on 2022 – with a wide range of differing contributions from one country to another, as demonstrated in Figure 7.1.


FIGURE 7.1: PV CONTRIBUTION TO ELECTRICITY DEMAND 2023


SOURCE IEA PVPS & OTHERS

FIGURE 7.2: SHARE OF RENEWABLE IN THE GLOBAL ELECTRICITY PRODUCTION IN 2023


SOURCE REN21, IEA PVPS

FIGURE 7.3: NEW RENEWABLE INSTALLED CAPACITY IN 2023


SOURCE REN21, GWEC, IHA, IEA PVPS

PV ELECTRICITY PRODUCTION / CONTINUED

TABLE 7.1: 2023 PV ELECTRICITY STATISTICS IN IEA PVPS COUNTRIES

COUNTRY	FINAL ELECTRICITY CONSUMPTION 2023	HABITANTS 2023	GDP 2023	SURFACE	AVERAGE YIELD	PV ANNUAL INSTALLED CAPACITY 2023	PV INSTALLED CAPACITY 2023	PV ELECTRICITY PRODUCTION	ANNUAL CAPACITY PER CAPITA	CUMULATIVE CAPACITY PER CAPITA	CUMULATIVE CAPACITY PER KM2	THEORETICAL PV PENETRATION
	TWH	MILLION	BUSD	KM2	KWH/KWP	MW	MW	TWH	W/CAP	W/CAP	KW/KM2	%
AUSTRALIA	273	27	1 724	7 690 000	1 400	4 153	34 522	48	156	1 296	4	17.7%
AUSTRIA	61	9	516	83 883	1 050	2 603	6 395	7	285	700	76	11.0%
BELGIUM	79	12	632	30 688	925	1 806	9 955	9	153	842	324	11.7%
CANADA	598	40	2 140	9 985 000	1 150	823	7 340	8	21	183	1	1.4%
CHINA	9 150	1 411	17 795	9 634 000	1 350	277 176	691 241	933	196	490	72	10.2%
DENMARK	36	6	404	44 000	975	487	3 910	4	82	658	89	10.6%
FINLAND	80	6	300	338 432	850	318	1 008	1	57	181	3	1.1%
FRANCE	446	68	3 031	551 500	1 100	3 961	23 664	26	58	347	43	5.8%
GERMANY	408	84	4 456	357 588	808	15 005	82 305	67	178	974	230	16.3%
ISRAEL	71	10	510	20 770	1 797	1 150	5 657	10	118	580	272	14.2%
ITALY	306	59	2 255	302 070	1 122	5 255	30 319	34	89	516	100	11.1%
JAPAN	851	125	4 213	377 975	1 100	6 300	91 366	101	51	734	242	11.8%
SOUTH KOREA	594	52	1 713	100 401	1 261	3 306	27 619	35	64	534	275	5.9%
MALAYSIA	171	34	400	330 621	1 314	2 956	3 390	4	21	99	10	2.6%
MOROCCO	35	38	141	710 850	1 750	728	1 714	3	19	45	2	8.5%
NETHERLANDS	108	18	1 118	41 500	994	4 788	23 037	23	268	1 288	555	21.3%
NORWAY	126	6	486	323 806	760	303	657	0	55	119	2	0.4%
PORTUGAL	51	11	287	92 225	1 407	1 285	3 822	5	122	363	41	10.6%
SPAIN	254	48	1 581	505 990	1 500	8 987	38 986	58	186	806	77	23.0%
SWEDEN	135	11	593	410 000	950	1 688	4 215	4	160	400	10	3.0%
SWITZERLAND	54	9	885	41 285	970	1 641	6 375	W	185	720	154	11.5%
SOUTH AFRICA	195	60	378	1 219 090	1 733	2 965	7 504	13	49	124	6	6.7%
THAILAND	214	72	515	1 219 092	1 522	4 587	9 002	14	64	125	7	6.4%
TÜRKIYE	327	85	1 108	783 560	1 471	1 867	14 393	21	22	169	18	6.5%
USA	4 252	335	27 361	9 147 282	1 500	33 884	177 344	266	101	530	19	6.3%
IEA PVPS	18 874	2 634	74 541	44 341 608	1 300	385 766	1 305 739	1 701	146	496	29	9.0%
BRAZIL	508	216	2 174	30 530	1 506	12 446	37 817	57	58	175	1 239	11.2%
INDIA	1 320	1 429	3 550	357 172	1 625	13 020	92 645	151	9	65	259	11.4%
NON IEA PVPS	6 884	5 391	30 894	89 983 827	1 300	70 244	336 218	433	13	62	4	6.3%
WORLD	25 758	8 025	105 435	134 325 435	1 300	456 010	1 641 956	2 135	57	205	12	8.3%

SOURCE IEA PVPS & OTHERS



Renewable energies contributed more than 30% to global electricity production in 2023, up 0.4% on 2022. Whilst a significant part of new capacity was solar, (around 75%), with a lower capacity factor than wind or hydro power and a year with excellent wind in several large wind markets, its actual share in annual electricity generation decreased slightly to approximately 5.5%.

PV INTEGRATION AND SECTOR COUPLING

THE ENERGY STORAGE MARKET

Energy storage can take different forms – when coupled with PV, the most common are batteries in standalone systems, generally found in residential systems far from the grid, however distributed residential batteries are increasingly used to improve self-consumption rates, whilst batteries are used to keep microgrids stable, providing both services and energy, and grid-scale batteries are slowly being deployed to provide system services.

In general, distributed battery storage is seen as an opportunity to solve grid integration issues linked to PV and to increase the self-consumption ratio of distributed PV plants to improve the payback time of PV systems. Despite their decreasing costs, such solutions are not yet economically viable in all countries and market segments – with some stand out countries such as Australia, China, Germany, Italy, and certain states in the USA. Around the world, the adoption of batteries is on the rise both in the residential segments and in the commercial segments as more and more consumers are willing to maximise their self-consumption and to optimize their consumption profile. In some countries, this is encouraged to relieve grid congestion or peak loads, either through subsidies (Australia, Austria, China, Spain, Japan, United States) or the terms attached to self-consumption policies such as time-of-use net billing and/or regulations on new building construction (both notably in use in California in the USA).

More large-scale PV plants are being built in combination with grid-scale batteries, which can be used to stabilize grid injection, reduce curtailment, and, in some cases, to provide ancillary services to the grid such as fast response voltage stability or peak power surges. New requirements for grid integration in tenders tend to favour the use of stationary batteries in utility-scale plants to smooth the output of the plant, reduce curtailment or reduce the need for grid capacity reinforcement, however this trend would require some more years to be confirmed.

Independently of PV systems, an increasing number of utility scale batteries are being installed around the world and their system size is increasing each year. The largest markets for utility scale batteries were China, USA, Europe (UK, Italy), Australia but also Japan and South Korea. In most of the markets, battery project developers are also developers of solar projects or energy majors. These grid-scale batteries provide services that would otherwise have been provided by turbine-based generation (coal, gas or hydro power), enabling higher PV penetration rates and a faster energy transition.

Globally, the largest part of batteries sold are used for transportation in EVs and stationary storage remains the exception with smaller volumes. However, the rapid development of electric mobility is driving battery prices down much faster than any could have expected in the stationary market alone, and is making storage increasingly an economically viable tool for increasing PV penetration rates.

THE ELECTRIFICATION OF TRANSPORT

The electrification of transport continues to advance around the world in parallel to the growth of solar. Charging EVs during peak load implies rethinking power generation, grid management and smart metering, and concepts such as virtual self-consumption could provide a framework for EVs as mobile storage for excess PV generation. In parallel, the use of transport infrastructure to support PV is opening up possibilities for future direct injection to transport networks (rail, tram) or for supplying vehicle charging stations (car ports and parking canopies, noise barriers).

THE ELECTRIFICATION OF HEATING AND COOLING

With the development of self-consumption, grid congestion and the need for peak shaving to relieve grid instability, the use of PV led heating and cooling is becoming increasingly common.

Heating with PV is mainly used to heat domestic hot water, as a way of increasing self-consumption, whether it be in response

From PV to VIPV and VAPV

With its distributed nature, PV fits perfectly with EV charging during the day when cars are stationed in commercial and office parking or at home. Such slow charging is also highly compatible with distribution grid constraints. Finally, the integration of PV in the vehicles themselves (VIPV), also offers opportunities to alleviate the burden on the grid, increase the autonomy of EVs, provide greater driver comfort and connects the automotive and PV sectors. The IEA PVPS Task 17 deals with this fast-emerging subject.

PV INTEGRATION AND SECTOR COUPLING / CONTINUED

to electricity consumption costs or grid capacity and congestion problems. Domestic hot water manufacturers now integrate devices to directly link extra PV production to the electric boiler. Specific recommendations exist for connection and metering of storage systems in Switzerland for instance.

In hot climates such as Australia, China, Japan, as well as the states of Florida and California (USA), PV has already been used to provide electricity for cooling for several years. However, as climate change intensifies and electricity costs go up whilst grid infrastructure becomes more fragile to heatwaves, the use of PV at the point of consumption to reduce grid loads whilst cooling needs are high is becoming more and more necessary. The electrification of heating and cooling through the increasing use of heat pumps will also increase demand for PV at the point of consumption.

For larger coupling, no real commercial products seem available. Nevertheless, designs of solar PV systems based on self-consumption are linked to some specific uses, for example of adapted water chillers including cold water storage. Commercial companies that intensively use cooling (especially in the food chain), and supermarkets are increasingly turning to PV installations on their buildings to reduce the electricity load of refrigerating units.

The use of solar energy (PV or thermal) for cooling is the subject of the IEA SHC Task 65 (<https://task65.iea-shc.org/>) that concluded in June 2024, which focused on innovative ways to adapt and develop existing technologies (solar and heat pumps) for sunny and hot climates.

GREEN HYDROGEN AND HYDROGEN DERIVATIVES

Green hydrogen refers to hydrogen produced from renewable sources, as opposed to hydrogen produced from fossil fuels or nuclear power. Hydrogen (or its derivatives such as ammonia) is increasingly seen as a partial answer to decarbonize some sectors such as heavy industry, agriculture, the maritime sector and long distance, heavy weight road transport. Early

industrialization projects around the creation of green hydrogen (or other molecules) continue to be funded each year. Hydrogen, or its derivatives, produced by competitive PV can also be stored and used to produce electricity later, even if the overall efficiency decreases significantly. It is seen as a way of increasing PV capacity to supply electricity grids when there is a need, but also to store excess production when grid needs are lower, even if technological challenges remain. A number of very large PV projects are in the planning stages to power hydrogen electrolyzers, particularly in the Middle East, China and Australia (see chapter 3 for policy developments and chapter 6).



ANNEXES

ANNEX 1: CUMULATIVE INSTALLED PV CAPACITY (MWP) FROM 1992 TO 2023

	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	
AUSTRALIA	7	9	11	13	16	19	23	25	29	34	39	46	52	61	70	82	105	187	571	1 376	2 416	3 226	4 092	5 109	5 985	7 132	11 586	16 399	21 091	26 129	30 368	34 522	
AUSTRIA	1	1	1	1	2	2	3	4	5	6	10	17	21	24	26	28	32	53	95	187	363	626	785	997	1 096	1 289	1 455	1 702	2 043	2 283	3 792	6 395	
BELGIUM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	24	112	671	1 108	2 176	2 870	3 139	3 245	3 355	3 535	3 865	4 310	5 127	6 273	7 123	8 149	9 955	
CANADA	1	1	2	2	3	3	4	6	7	9	10	12	14	17	20	26	33	55	281	559	766	1 211	1 843	2 519	2 665	2 913	3 130	3 388	3 713	5 752	6 517	7 340	
CHINA	0	0	0	0	0	0	0	0	11	16	34	44	54	62	72	92	132	292	792	3 492	6 692	17 682	28 322	43 472	78 022	130 882	175 142	205 440	253 640	308 520	414 065	649 034	
DENMARK	0	0	0	0	0	0	1	1	1	1	1	2	2	2	2	3	3	4	7	29	499	698	751	979	1 061	1 139	1 254	1 362	1 626	1 850	3 423	3 910	
FINLAND	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	2	2	3	5	7	9	9	9	9	20	37	80	134	214	313	417	691	
FRANCE	2	2	2	3	4	6	8	9	11	14	17	21	24	26	38	76	218	440	1 446	3 562	4 905	5 692	6 837	7 920	8 656	9 713	10 755	11 930	13 008	16 737	19 703	23 664	
GERMANY	6	9	12	18	28	42	54	70	114	176	296	435	1 105	2 056	2 899	4 170	6 120	10 566	18 006	25 916	34 077	36 710	37 900	39 224	40 679	42 293	45 181	49 016	53 901	60 107	67 300	82 305	
ISRAEL	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	22	67	186	272	377	588	771	877	952	1 358	1 960	2 414	3 349	4 507	5 657	
ITALY	8	12	14	16	16	17	18	18	19	20	22	26	31	37	50	100	496	1 277	3 605	13 141	16 736	18 198	18 607	18 915	19 297	19 682	20 108	20 885	21 650	22 594	25 064	30 319	
JAPAN	19	24	31	43	60	91	133	209	330	453	637	880	1 132	1 422	1 708	1 919	2 144	2 627	3 618	4 914	6 632	13 999	23 339	34 151	42 040	49 500	56 162	63 192	71 888	78 413	85 066	91 366	
MALAYSIA	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	2	4	34	145	213	273	352	401	793	1 292	1 800	2 308	2 680	3 380	
MOROCCO	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1 714	
NETHERLANDS	0	0	0	0	0	1	1	1	5	9	16	22	40	43	45	49	59	69	111	170	390	767	1 069	1 536	2 061	2 914	4 609	7 225	10 717	14 349	18 249	23 037	
NORWAY	0	0	0	0	0	0	0	0	0	6	6	6	6	7	7	7	8	8	8	9	9	10	12	14	26	44	68	120	160	201	354	657	
PORTUGAL	0	0	0	0	0	0	0	0	0	1	2	2	3	3	3	18	68	108	134	175	244	299	418	454	519	585	673	907	1 077	1 647	2 537	3 822	
SOUTH AFRICA	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	311	1 393	1 487	2 077	2 146	2 206	2 669	3 969	4 427	4 539	7 504	
SOUTH KOREA	0	0	0	0	0	0	0	0	0	0	5	6	9	14	36	81	357	524	650	729	1 024	1 555	2 481	3 615	4 502	5 855	8 099	12 665	17 323	21 199	24 313	27 619	
SPAIN	0	0	0	0	0	0	0	0	0	3	6	14	25	52	150	742	4 021	4 070	4 595	5 080	5 439	5 566	5 641	5 705	5 778	5 928	6 306	11 628	15 875	21 513	29 999	38 986	
SWEDEN	1	1	1	2	2	2	2	3	3	3	3	4	4	4	5	6	8	9	11	15	23	42	77	125	184	269	429	707	1 107	1 606	2 527	4 215	
SWITZERLAND	5	6	7	8	10	11	13	14	16	18	20	22	24	28	30	37	49	80	125	223	437	756	1 061	1 394	1 664	1 906	2 173	2 498	2 973	3 651	4 734	6 375	
THAILAND	0	0	0	0	0	0	0	0	0	0	0	0	0	0	25	32	32	42	185	379	524	961	1 436	1 557	2 584	3 194	3 650	3 666	3 715	4 215	4 415	9 002	
TURKIYE	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	6	32	64	358	1 175	4 206	7 339	8 551	9 424	10 917	12 526	14 393	
USA	0	0	0	0	0	0	0	0	0	0	0	0	0	0	111	190	295	455	753	1 188	2 017	3 597	7 130	12 076	18 321	25 821	40 973	51 818	62 498	76 274	95 550	120 388	143 469
REST OF EU COUNTRIES	0	0	0	0	0	0	0	0	0	0	1	11	21	23	34	42	107	674	2 659	3 759	5 962	8 404	8 815	9 167	9 571	9 943	10 753	12 363	16 617	22 856	32 200	46 852	
TOTAL IEA PVPS	50	65	90	115	150	206	271	372	572	784	1 143	1 566	2 688	4 116	5 552	8 013	14 884	23 036	40 132	70 067	97 580	132 214	167 718	209 701	276 832	360 391	445 838	529 088	642 200	778 034	968 976	1 329 105	
BRAZIL	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	26	38	53	66	86	93	1 160	2 416	4 697	8 474	14 504	25 371	37 817	
INDIA	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	57	272	1 299	2 384	3 249	5 436	9 547	22 561	33 363	43 431	47 826	61 490	79 625	92 465	
TOTAL NON IEA PVPS	0	0	0	0	0	0	0	0	1	8	14	24	34	46	67	94	139	249	576	2 304	4 781	8 049	12 800	21 588	31 076	50 086	69 006	99 291	131 515	189 087	216 970	312 652	
TOTAL	50	65	90	115	150	206	271	372	573	791	1 157	1 591	2 732	4 162	5 619	8 107	15 023	23 285	40 708	72 371	102 361	140 263	180 518	231 288	307 908	411 016	514 844	628 393	773 515	947 180	1 185 946	1 641 956	

SOURCE IEA PVPS 8 OTHERS

ANNEXES / CONTINUED

ANNEX 2: ANNUAL INSTALLED PV CAPACITY (MW) FROM 1992 TO 2023

	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	
AUSTRALIA	7	2	2	2	3	3	4	3	4	4	6	6	7	8	10	12	22	83	388	806	1 039	811	866	1 018	876	1 147	4 454	4 813	4 692	5 038	4 239	4 153	
AUSTRIA	1	0	0	0	0	0	0	1	1	1	4	6	4	3	2	2	5	20	43	92	176	263	159	152	159	173	186	247	341	739	1 009	2 603	
BELGIUM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	18	88	559	437	1 068	694	269	106	110	180	330	445	817	1 146	850	917	1 805	
CANADA	1	0	0	0	1	1	1	1	1	2	1	2	2	3	4	5	7	62	187	277	208	445	633	675	146	249	217	258	325	2 038	765	823	
CHINA	0	0	0	0	0	0	0	0	0	11	5	19	10	8	10	20	40	160	500	2 700	3 200	10 990	10 640	15 150	34 550	52 860	44 260	30 300	48 200	54 880	105 545	234 969	
DENMARK	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	3	22	470	199	53	228	81	78	115	109	264	718	1 754	487		
FINLAND	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	2	2	2	0	0	0	11	17	43	53	81	98	104	269	318	
FRANCE	2	0	0	1	2	2	2	2	2	3	3	4	3	2	12	38	143	222	1 006	2 116	1 344	786	1 145	1 083	716	1 077	1 042	1 175	1 168	3 639	2 966	3 961	
GERMANY	6	3	3	6	10	14	12	16	44	62	120	139	670	951	843	1 271	1 950	4 446	7 440	7 910	8 161	2 633	1 190	1 324	1 455	1 614	2 888	3 835	4 885	6 206	7 193	15 005	
ISRAEL	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	21	45	119	86	105	211	183	106	75	406	602	454	935	1 158	1 150		
ITALY	8	4	2	2	2	0	1	1	1	1	2	4	5	7	13	50	396	781	2 328	9 536	3 655	1 402	409	308	382	385	426	758	785	944	2 470	5 255	
JAPAN	19	5	7	12	16	32	42	75	122	123	184	223	272	290	287	210	225	483	991	1 296	1 718	6 968	9 740	10 811	7 889	7 460	6 662	7 030	8 676	6 545	6 653	6 300	
MALAYSIA	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2	34	111	67	61	78	49	517	499	508	372	1 068	710		
MOROCCO	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	493	287	728
NETHERLANDS	0	0	0	0	0	0	0	0	0	4	3	8	6	18	4	2	3	10	42	59	220	377	302	467	525	853	1 656	2 616	3 492	3 632	4 200	4 788	
NORWAY	0	0	0	0	0	0	0	0	0	6	0	0	0	0	0	1	0	0	0	0	1	1	2	2	11	18	25	51	40	41	153	303	
PORTUGAL	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	14	50	40	26	41	69	55	119	36	66	88	234	170	571	999	1 285		
SOUTH AFRICA	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	305	1 081	94	590	69	60	463	1 300	458	112	2 965		
SOUTH KOREA	0	0	0	0	0	0	0	0	0	0	5	1	3	5	22	45	276	167	127	79	295	531	926	1 134	887	1 333	2 265	4 566	4 658	3 876	3 278	3 306	
SPAIN	0	0	0	0	0	0	0	0	0	3	3	8	11	27	98	592	3 279	49	525	485	359	127	75	65	72	150	378	5 322	4 247	5 639	8 495	8 987	
SWEDEN	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	2	1	2	4	8	19	35	48	59	85	160	291	400	499	921	1 688	
SWITZERLAND	5	1	1	1	2	2	2	2	2	2	2	2	2	4	2	7	12	30	46	98	214	319	305	333	270	242	267	325	475	677	1 084	1 641	
THAILAND	0	0	0	0	0	0	0	0	0	0	0	0	0	25	7	0	0	10	6	194	145	437	475	122	1 027	610	456	16	49	500	200	4 587	
TURKIYE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1 867		
USA	0	0	0	0	0	0	0	0	0	0	0	0	111	79	105	160	298	435	829	1 920	3 193	4 946	6 245	7 500	15 152	10 845	10 680	13 776	19 276	24 819	23 101	33 875	
REST OF EU COUNTRIES	0	0	0	0	0	0	0	0	0	0	0	10	10	2	11	7	65	588	1 984	1 100	2 204	2 441	412	352	405	372	792	1 634	4 248	6 239	8 935	13 652	
TOTAL IEA PVPS	50	16	25	25	35	56	65	101	199	212	360	423	1 131	1 419	1 436	2 460	6 871	8 153	16 958	29 935	27 517	34 634	35 503	41 983	67 131	84 099	85 052	83 245	113 133	136 252	191 779	330 128	
BRAZIL	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	26	12	15	14	20	7	1 067	1 256	2 281	3 777	6 000	10 867	12 446	
INDIA	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	57	214	1 027	1 085	865	2 186	4 111	13 013	10 683	10 068	4 395	13 664	18 135	13 020	
TOTAL NON IEA PVPS	0	0	0	0	0	0	0	0	1	7	6	10	10	12	21	27	45	109	326	1 728	2 475	3 268	4 751	8 788	9 478	19 009	24 160	30 286	32 225	37 572	49 897	95 882	
TOTAL	50	16	25	25	35	56	65	101	200	219	366	433	1 141	1 430	1 457	2 487	6 916	8 262	17 285	31 663	29 991	37 902	40 254	50 770	76 608	103 108	109 192	113 531	145 358	173 824	241 676	456 010	



ANNEXES / CONTINUED

ANNEX 3: CURRENCY EXCHANGE RATES

COUNTRY	CURRENCY CODE	EXCHANGE RATE
		IN 2023(1 USD =)
AUSTRALIA	AUD	1.506
BRAZIL	BRL	4.994
CANADA	CAD	1.35
CHILE	CLP	952.43
CHINA	CNY	7.075
DENMARK	DKK	6.89
EUROZONE	EUR	0.924
INDIA	INR	82.572
ISRAEL	ILS	3.687
JAPAN	JPY	140.511
SOUTH KOREA	KRW	1338.93
MALAYSIA	MYR	4.61
MEXICO	MXN	17.733
MOROCCO	MAD	10.275
NORWAY	NOK	10.564
SOUTH AFRICA	ZAR	18.457
SWEDEN	SEK	10.613
SWITZERLAND	CHF	0.899
THAILAND	THB	34.802
TÜRKIYE	TRY	23.824
USA	USD	1

ANNEXES / CONTINUED

ANNEX 4: COUNTRY AND MARKET GROUPINGS

COUNTRY/MARKET	SUB REGION	REGION
ALGERIA	AFRICA	MIDDLE EAST AND AFRICA
ARGENTINA	LATIN AMERICA	THE AMERICAS
AUSTRALIA	OCEANIA	ASIA PACIFIC
AUSTRIA	EUROPE	EUROPE
BELGIUM	EUROPE	EUROPE
BRAZIL	LATIN AMERICA	THE AMERICAS
BULGARIA	EUROPE	EUROPE
CANADA	NORTH AMERICA	THE AMERICAS
CHILE	LATIN AMERICA	THE AMERICAS
CHINA	ASIA	ASIA PACIFIC
CHINESE TAIPEI	ASIA	ASIA PACIFIC
COLOMBIA	LATIN AMERICA	THE AMERICAS
CROATIA	EUROPE	EUROPE
CYPRUS	EUROPE	EUROPE
CZECH REPUBLIC	EUROPE	EUROPE
DENMARK	EUROPE	EUROPE
ECUADOR	LATIN AMERICA	THE AMERICAS
EGYPT	AFRICA	MIDDLE EAST AND AFRICA
ESTONIA	EUROPE	EUROPE
FINLAND	EUROPE	EUROPE
FRANCE	EUROPE	EUROPE
GERMANY	EUROPE	EUROPE
GREECE	EUROPE	EUROPE
HONDURAS	CENTRAL AMERICA + CARIBBEAN	THE AMERICAS
HUNGARY	EUROPE	EUROPE
ICELAND	EUROPE	EUROPE
INDIA	ASIA	ASIA PACIFIC
INDONESIA	ASIA	ASIA PACIFIC
IRELAND	EUROPE	EUROPE
ISRAEL	MIDDLE EAST	MIDDLE EAST AND AFRICA

COUNTRY/MARKET	SUB REGION	REGION
ITALY	EUROPE	EUROPE
JAPAN	ASIA	ASIA PACIFIC
JORDAN	MIDDLE EAST	MIDDLE EAST AND AFRICA
KAZAKHSTAN	ASIA	ASIA PACIFIC
SOUTH KOREA	ASIA	ASIA PACIFIC
LATVIA	EUROPE	EUROPE
LITHUANIA	EUROPE	EUROPE
LUXEMBOURG	EUROPE	EUROPE
MALAYSIA	ASIA	ASIA PACIFIC
MALTA	EUROPE	EUROPE
MEXICO	NORTH AMERICA	THE AMERICAS
MOROCCO	AFRICA	MIDDLE EAST AND AFRICA
NETHERLANDS	EUROPE	EUROPE
NORWAY	EUROPE	EUROPE
PAKISTAN	ASIA	ASIA PACIFIC
PANAMA	CENTRAL AMERICA + CARIBBEAN	THE AMERICAS
PERU	LATIN AMERICA	THE AMERICAS
PHILIPPINES	ASIA	ASIA PACIFIC
POLAND	EUROPE	EUROPE
PORTUGAL	EUROPE	EUROPE
ROMANIA	EUROPE	EUROPE
RUSSIA	EURASIA	EUROPE
SAUDI ARABIA	MIDDLE EAST	MIDDLE EAST AND AFRICA
SERBIA	EUROPE	EUROPE
SINGAPORE	ASIA	ASIA PACIFIC
SLOVAKIA	EUROPE	EUROPE
SLOVENIA	EUROPE	EUROPE
SOUTH AFRICA	AFRICA	MIDDLE EAST AND AFRICA
SPAIN	EUROPE	EUROPE
SWEDEN	EUROPE	EUROPE



ANNEXES / CONTINUED

ANNEXE 4: COUNTRY AND MARKET GROUPINGS

COUNTRY/MARKET	SUB REGION	REGION
SWITZERLAND	EUROPE	EUROPE
THAILAND	ASIA	ASIA PACIFIC
TÜRKIYE	EURASIA	EUROPE
UK	EUROPE	EUROPE
UKRAINE	EUROPE	EUROPE
UNITED ARAB EMIRATES	MIDDLE EAST	MIDDLE EAST AND AFRICA
USA	NORTH AMERICA	THE AMERICAS
VIETNAM	ASIA	ASIA PACIFIC

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